

Infections and Malignancies in the Kidney Transplant Recipient

Phuong-Thu T. Pham, Joanna Schaenman, Phuong-Chi T. Pham

INFECTIOUS DISEASES

Infection follows cardiovascular (CV) disease as the second most common cause of death with a functioning graft in kidney transplant recipients. Infection and malignancy are more commonly the cause of death among those without diabetes. Predisposing risk factors for infectious complications include immunosuppression, increased recipient age, indwelling Foley or vascular catheters, surgical drains, drug-induced leukopenia, and metabolic derangement. Epidemiologic exposures, shifts in nosocomial flora associated with repeated antimicrobial exposures (antimicrobial resistance), and improvement in molecular diagnostic assays also contribute to the emergence of novel infections and changes in the epidemiology of infections.¹

Immunizations Before and After Transplantation

All kidney transplant candidates should receive immunization for hepatitis B, pneumococcus, and other standard immunizations appropriate for age and the presence of end-stage kidney disease. Inactivated vaccine should be completed by 2 weeks prior to transplantation to optimize immune response to vaccination, and a minimum of 4 weeks should elapse between live attenuated virus vaccine administration and transplantation to minimize the possibility of live vaccine-derived infection in the posttransplantation period.² Live virus or live organism vaccines should be avoided after transplantation. Immunizations using inactivated or killed microorganisms, components, and recombinant moieties are safe for transplant recipients. Most centers restart vaccinations 3 to 6 months after transplantation. Vaccination during the first 3 months after transplantation may result in suboptimal response and protection because of increased levels of immunosuppression early posttransplant, especially in patients who receive lymphocyte-depleting agents as part of induction. Ensuring adequate response to hepatitis B vaccination is important to prevent transmission of donor-derived infection. Seasonal influenza vaccine is safe and effective, and no conclusive evidence exists for a link between vaccination and allograft dysfunction. Household members, close contacts, and health care workers should also be fully immunized. In addition, they should receive annual influenza vaccine, preferably with inactivated influenza vaccine. If live attenuated vaccine is used, frequent handwashing for a 2-week period after vaccination is recommended. With the exception of smallpox and oral polio vaccines, there is little to no risk to the transplant recipient if family members or close contacts receive live vaccines. Pets should also be fully immunized. Immunization of pets with live vaccines (e.g., canine *Bordetella bronchiseptica* intranasal vaccine) poses little to no risk of transmission.² Recommended immunizations before and after transplantation are listed in Table 110.1. For vaccination of transplant recipients who travel to countries where endemic infections such as malaria are present, see Buchan et al.³ All solid organ transplant recipients should be seen by a travel medicine specialist prior to traveling to destinations with higher rates of infection.³

Infectious Causes

The type and occurrence of infections in the immunocompromised transplant recipient follow a predictable pattern based on time since transplantation (Table 110.2). Timing of infections may be altered by the intensity of immune suppression, use of target antibiotic prophylaxis, and patient exposures.

Risk Factors for Posttransplant Infectious Complications

Risk factors for posttransplant infectious complications include active or latent infections in the donors, net state of immunosuppression, surgical instrumentation, wound-healing issues, abdominal fluid collections, reactivation of latent infections, epidemiologic exposure, metabolic conditions (e.g., diabetes, uremia), infections with immunomodulating viruses, and hypogammaglobulinemia, among others. Severe hypogammaglobulinemia (immunoglobulin [Ig] G < 400 mg/dL) during the first year posttransplantation is associated with a significant increase in the risk for cytomegalovirus (CMV) infection, fungal and respiratory infections, and 1-year all-cause mortality among recipients of solid organ transplantation.⁴ Although controlled trials are lacking, posttransplant monitoring of IgG levels and Ig replacement therapy to keep IgG level above 700 to 800 mg/dL may reduce infection rates in patients with hypogammaglobulinemia.

Donor-Derived Infections

Donor-derived infections include bacterial, fungal (*Candida* spp.), viral (human immunodeficiency virus [HIV], hepatitis B and C [HBV and HCV], cytomegalovirus [CMV]), and parasitic infections (malaria, *Babesia* and *Balamuthia* infections, amebic meningitis). Allograft-transmitted infections with unusual viruses include lymphocytic choriomeningitis virus, West Nile virus, and rabies. Despite routine surgical prophylaxis, transmission of resistant organisms, including methicillin-resistant *Staphylococcus aureus* (MRSA), vancomycin-resistant enterococci (VRE), and azole-resistant *Candida* spp. may still occur. Donor-derived carbapenem-resistant Enterobacteriaceae (CRE) infections are associated with high morbidity and mortality. Donation of organs that are directly involved in CRE infection should be evaluated on a case-by-case basis with input from an infectious disease consultation given the high risk of donor-derived infection (e.g., kidneys from donors with CRE urinary tract infection [UTI], donors with CRE bacteremia). Systematic review and meta-analyses revealed that donor-derived viral infections were most prevalent (56.0%), followed by bacterial (15.5%), fungal (15.5%), and parasitic (13.0%) infections. Among viral donor-derived transmissions, HIV and human T-cell lymphotropic virus (HTLV) were the most frequently observed infections (9.7%), followed by West Nile virus (6.3%). Most commonly reported donor-transmitted bacterial infections were *Mycobacterium tuberculosis* (4.8%) and *Pseudomonas aeruginosa* (4.3%); *Candida* spp.

TABLE 110.1 Recommended Immunizations Before and After Transplantation

Vaccine	Inactivated (I) Virus or Live Attenuated (LA) Virus	Before Transplant	After Transplant (>3–6 Mo Posttransplant)	Comments
MMR	LA	Yes	Contraindicated	Presence of measles antibody can be assessed to determine need for vaccination or revaccination prior to transplantation
DPT	I	Yes	See comments	Diphtheria and tetanus: booster every 10 yr
Herpes zoster (Shingrix)	I	Yes	Yes	Indicated for age ≥ 50 yr
Poliovirus	I	Yes	Yes	For travelers to endemic areas (e.g., some parts of Asia, Africa)
<i>Haemophilus influenza</i> type b	I	Yes	Yes	Indicated for patients with sickle cell disease, complement deficiency, or those who had a splenectomy
Influenza A and B ^a	I	Yes	Yes	Annually ^b All patients who are >3 mo posttransplant should receive influenza vaccine May be administered in the immediate posttransplant period during an outbreak
Hepatitis A	I	Yes	Yes	Recommended posttransplant if not administered pretransplant For travelers to endemic areas
Hepatitis B (Hepatisav, Engerix)	I	Yes	Yes	Recommended posttransplant if not administered pretransplant Monitor titers for response Repeat series if needed
PCV20 (Pevnar)	I	Yes	Yes	If possible, PCV13 should be administered before PPSV23
PPSV23 (Pneumovax)	I	Yes	Yes	For adults who have not previously received PPSV23, administer PCV13, followed by PPSV23 at least 8 wk later For adults who just received PPSV23 but have not received PCV13, wait 12 mo before administering PCV13 Adults <65 years who are high risk should receive 2 doses of PPSV23 5 yr apart (but no sooner than 8 wk after PCV13), with a third dose after they turn age 65 yr
HPV	I	Yes	Yes	Nonpregnant female candidates ages 11–26 yr, males ages 11–21 yr
<i>Neisseria meningitidis</i> (both quadrivalent and group B vaccines)	I	Yes	Yes	Recommended for patients with properdin terminal component deficiencies or receiving eculizumab therapy, or those with functional or anatomic asplenia Others: military members travelers to high-risk areas, college freshmen living on campus
COVID-19	Vaccine type ^c : mRNA Replication defective adenovirus vector ^d	Yes	Yes	Preexposure prophylaxis with tixagevimab plus cilgavimab is recommended for all immunosuppressed kidney transplant recipients unless contraindicated (must complete primary vaccine series) Monovalent vaccines protect against the original SARS-CoV-2 whereas bivalent vaccines protect against both the original and the Omicron variants BA.4 and BA.5 (should be administered at least 2 months after completion of the primary series)

^aLive attenuated nasal influenza vaccine has been removed from the US market.

^bIf possible, patients being treated for or recently treated for allograft rejection should avoid vaccinations until 3 months after treatment to ensure optimal immune response. However, vaccination should not be delayed if such delay represents more of a risk than the risk of impaired response. All transplant recipients over age 65 years should receive the high-dose vaccine where available.

^cVaccines available in the United States: mRNA (Pfizer-BioNTech, Moderna), protein-based (Novavax), replication defective adenovirus vector (Johnson & Johnson's Janssen).

^dThe US Centers for Disease Control and Prevention recommend that the Janssen vaccine only be considered in certain situations due to safety concerns (last updated August 2022).

AST, American Society of Transplantation; DPT, diphtheria, pertussis, tetanus; HPV, human papillomavirus; MMR, measles, mumps, rubella; PCV, pneumococcal conjugate vaccine; PPSV, pneumococcal polysaccharide vaccine, SARS-CoV-2, severe acute respiratory syndrome coronavirus 2.

(3.9%) and *Toxoplasma gondii* (3.4%) were the most common donor-derived fungal and parasitic infections, respectively.⁵ Nucleic acid testing (NAT) for HIV and hepatitis infections should be performed on all donors, and all recipients should undergo posttransplant infection surveillance. If a donor is at increased risk for HIV, HBV, or HCV

transmission, potential recipients of such kidneys should undergo additional counseling prior to organ acceptance (Box 110.1). Currently all deceased and living organ donors and potential recipients are tested for the novel COVID-19 by polymerase chain reaction (PCR) prior to organ procurement or transplantation. One fatal case of transmission

TABLE 110.2 Timetable of Infections

Timetable of Infections	Comments
First Month After Transplant Bacterial (Sites and Sources) • Urinary tract • Respiratory system • Bacteremia • Surgical wound or intraabdominal sources (lymphoceles, hematoma, urine leak) • Vascular access or instrumentations (catheters, drains, urinary stents) • Anatomic or functional genitourinary tract abnormalities (ureteral stricture, vesicoureteric reflux, neurogenic bladder) • <i>Clostridioides difficile</i> or center-specific multidrug-resistant species Viral • Uncommon except HSV Fungal • <i>Candida</i> spp. predominate (recipient pretransplant colonization or donor derived)	• Common nosocomial bacterial pathogens and <i>Candida</i> spp. predominate • Risk can be mitigated by appropriate prophylaxis • Minimize or avoid environmental exposure at all times after transplant (primarily avoiding pigeons and areas of active building construction)
Organisms Transmitted With Donor Organs Months 1–6 Viral • CMV, HSV, VZV, EBV, HBV, HCV, BK virus (exogenous infection or reactivation of latent disease as a result of immunosuppression) • Others: HHV-6, HHV-7, influenza, parainfluenza, RSV, adenovirus Fungal • <i>Aspergillus</i> spp., <i>Cryptococcus</i>, agents of mucormycosis Bacterial • Recurrent urinary tract infections or pyelonephritis, <i>Nocardia</i>, <i>Listeria</i>, <i>Mycobacterium</i> spp. (tuberculous and nontuberculous), <i>Legionella</i> Parasitic • <i>Pneumocystis jirovecii</i>, <i>Toxoplasma</i> and <i>Strongyloides</i> spp., leishmaniasis, <i>Trypanosoma cruzi</i>	• Unconventional or opportunistic infections secondary to immunosuppression • Measures to reduce HBV and HCV infection and/or reactivation • Universal immunization for HBV • All patients with HBV infection (HBsAg or HBeAg positive) should receive lifelong antiviral prophylaxis^a • HCV patients should be referred to a hepatologist for consideration of DAA therapy^b • Minimize or avoid environmental exposure at all times after transplant
After 6 Months Stable Patients on Low-Dose Immunosuppressants • Community-acquired respiratory and GI viral pathogens History of Multiple Rejection Episodes Requiring Intensification of Immunosuppression • Viral infections (invasive CMV such as CMV colitis or pneumonitis, VZV, parvovirus B19), late opportunistic infections (<i>Pneumocystis</i>, <i>Cryptococcus</i>, <i>Listeria</i>, <i>nocardiosis</i>), tuberculosis Persistent Infections • HBV, HCV, papillomavirus, BK virus Geographically Restricted Infections • Examples: coccidioidomycosis, histoplasmosis, blastomycosis, paracoccidioidomycosis Deep-Seated Infections • Examples: osteomyelitis, paravertebral abscess • Predisposing risk factors: chronic skin infections, long-standing poorly controlled diabetes, peripheral vascular disease Associated With Malignancies • EBV (PTLD), papilloma virus (squamous cell carcinoma), HSV (cervical cancer), HHV-8 (Kaposi sarcoma)	• Infection risks associated with duration and intensity of immunosuppression and epidemiologic exposures • Excellent sustained virologic response achieved with direct-acting antiviral agents may lower the long-term incidence of chronic liver disease and cirrhosis in HCV patients • Minimize or avoid environmental exposure at all times after transplant

^aAll patients should receive antiviral therapy after transplantation. Transplant candidates with active HBV replication (i.e., detectable HBV DNA) should start therapy pretransplant.

^bIn patients with HBV and HCV coinfection, HBV antiviral prophylaxis should be considered prior to DAA therapy because HBV reactivation can occur as a consequence of DAA therapy. Patients with evidence of resolved infection (hepatitis B core antibody positive) should be monitored for HBV reactivation with serial HBV DNA and liver function tests during DAA therapy.

CMV, Cytomegalovirus; DAA, direct-acting antiviral agent; EBV, Epstein-Barr virus; GI, gastrointestinal; HBeAG, hepatitis B e-antigen; HBsAG, hepatitis B surface antigen; HBV, hepatitis B virus; HCV, hepatitis C virus; HHV, human herpesvirus; HSV, herpes simplex virus; PTLD, posttransplant lymphoproliferative disease; RSV, respiratory syncytial virus; VZV, varicella zoster virus.

BOX 110.1 United Network for Organ Sharing Protocol Surveillance for All Transplant Recipients

Pretransplant

At baseline:

- Any CDC-approved HIV testing modality
- HBsAg
- HCV quantitative PCR

Posttransplant

At 4–8 wk posttransplant:

- HIV RNA quantitative PCR
- HBV DNA quantitative PCR
- HCV RNA quantitative PCR

At 12 mo posttransplant (for liver transplant recipients only)

- HBsAg or HBV NAT, anti-HBsAb, and anti-HBcAb

CDC, Centers for Disease Control and Prevention; HBV, hepatitis B virus; HBcAb, hepatitis B core antibody; HBsAb, hepatitis B surface; HBsAg, hepatitis B surface antigen; HCV, hepatitis C virus; NAT, nucleic acid testing; PCR, polymerase chain reaction.

of COVID-19 from lung donor to recipient has been reported despite a negative nasopharyngeal swab.⁶ Hence, for thoracic organ transplantation, COVID-19 testing of the donor lower respiratory tract is recommended (if safe and feasible). Limited data suggest that transplantation of nonlung organs from COVID-19 infected donors can be safely performed. Assessing viral load using cycle threshold might be helpful in clinical decision-making (opinion based).

Month 1 After Transplantation

In the first month after transplantation, donor- and recipient-derived infections with common nosocomial bacterial microorganisms and *Candida* spp. predominate. Infections caused by multidrug-resistant bacteria are center specific. Despite the advent of potent broad-spectrum antimicrobial agents, infections caused by multidrug-resistant bacteria, including gram-positive bacteria (MRSA, VRE), extended-spectrum β -lactamases, and gram-negative bacilli and multidrug-resistant nonfermenting gram-negative bacilli, including CRE, continue to be important causes of morbidity and mortality.

Most bacterial infections during this period involve wounds, catheters, and drainage sites. Aspiration pneumonia and UTIs are common. Most UTIs are caused by gram-negative bacteria with *Escherichia coli* being the predominant causative organism. *Enterococcus* species account for the majority of gram-positive pathogens. Preventive measures for UTIs include early urethral catheter removal and antibiotic prophylaxis. Trimethoprim-sulfamethoxazole (TMP-SMX) prophylaxis during the first 3 to 6 months after transplantation may reduce the risk for UTIs and eliminate urosepsis unless anatomic or functional derangement of the urinary tract is present. Strict aseptic surgical techniques and perioperative use of first-generation cephalosporins reduce the incidence of wound infections. Except for herpes simplex virus (HSV), viral infections are uncommon during this period.

Months 1 to 6

During months 1 to 6, unconventional or opportunistic infections secondary to immunosuppression are most common. Viral infections such as CMV, HSV, varicella-zoster virus (VZV), Epstein-Barr virus (EBV), HBV, and HCV may occur from exogenous infection or reactivation of latent disease. Historically, reactivation of de novo HCV infection was an important cause of morbidity and mortality after transplantation. The advent of fixed-dose direct-acting antiviral

agent (DAA) combination therapy has improved outcomes in hepatitis C–positive transplant recipients. Repeated courses of antibiotics and corticosteroids increase the risk for fungal infections, whereas viral infections not only may result from immunosuppression but may further impair immunity and increase the risk for additional opportunistic infections. TMP-SMX prophylaxis eliminates or reduces the incidence of *Pneumocystis*, *Listeria monocytogenes*, *Nocardia* spp., and *Toxoplasma gondii* infection. Reactivation of latent infection such as from *Mycobacterium tuberculosis*, *Trypanosoma cruzi*, *Leishmania* spp., *Strongyloides stercoralis*, *Cryptococcus neoformans*, *Histoplasma capsulatum*, and *Coccidioides* and *Paracoccidioides* spp. may be observed. Community-acquired respiratory viruses are a common hazard in these vulnerable immunocompromised patients during this period, whereas BK polyomavirus infection is an important cause of allograft loss (discussed later).

After 6 Months

After 6 months, the infection risk is largely a function of chronic maintenance immunosuppression, exposure to T-cell-depleting agents, graft function, and epidemiologic exposures. Patients can be divided into three categories of infection risk.

The first category consists of most transplant recipients (70%–80%), who have satisfactory or good allograft function, relatively low doses of immunosuppressants, and no history of chronic viral infection. The risk for infection approaches that of the general population, with community-acquired respiratory viruses constituting the major infective agents. Opportunistic infections are unusual unless environmental exposure has occurred.

The second group (~10% of patients) consists of those with chronic viral infection that may include HBV, HCV, CMV, EBV, BK polyomavirus (BKPyV), or papillomavirus. In the setting of immunosuppression, such viral infections may lead to the development of BKPyV-associated nephropathy (BKVAN), posttransplantation lymphoproliferative disorder (PTLD), squamous cell carcinoma (papillomavirus), or progressive liver disease or cirrhosis (HBV, HCV). The sustained virologic response achieved with DAAs in both the dialysis and kidney transplant populations with chronic hepatitis C may lower the long-term incidence of chronic liver disease and cirrhosis in these populations. The advent of pangenotypic (genotypes 1–6) DAAs along with universal immunization for HBV can potentially eradicate HBV and HCV infection after transplant.^{7,8} New infections occurring after the first 6 months posttransplantation often reflect new exposures (e.g., *L. monocytogenes* [dietary ingestion], Lyme disease [tick exposure], and malaria [travelers to endemic areas]).¹ Alemtuzumab (a humanized monoclonal antibody directed against CD52 found on T and B cells) is associated with increased risk for late invasive viral and fungal infections because of its profound and prolonged effect on pan-T-cell depletion.

The third group (~10% of patients) consists of those who experience multiple rejection episodes requiring repeated exposure to heavy immunosuppression. These patients are the most likely to develop chronic viral infections and superinfection with opportunistic infections. Causative opportunistic pathogens include *Pneumocystis*, *Listeria*, *Nocardia*, *Cryptococcus*, and geographically restricted mycoses (coccidioidomycosis, histoplasmosis, blastomycosis, paracoccidioidomycosis). Risk factors include intensive immunosuppression for graft rejection, CMV infection, higher-dose corticosteroid therapy (defined as >20 mg of prednisone daily for at least 2 weeks), prolonged neutropenia, or flare of autoimmune disease. The widespread use of prophylactic therapy has resulted in a decrease in the incidence of *Pneumocystis* infection in solid organ transplant recipients with an estimated incidence of 0.3% to 1.5% compared with 3% to 5% historically.⁹

TABLE 110.3 Posttransplantation Antimicrobial Prophylaxis

Prophylaxis	Regimen	Comments
<i>Pneumocystis jirovecii</i>	First line: TMP-SMX $\times$ 6–12 mo ^a (lifelong in some, especially thoracic organs) Second line (sulfa allergies) ^b : dapsone, atovaquone, or aerosolized pentamidine ^c	TMP-SMX also reduces the incidence of <i>Toxoplasma gondii</i> , <i>Listeria monocytogenes</i> , and <i>Nocardia asteroides</i> and reduces the incidence of UTI in kidney transplant recipients Check G6PD before initiation of dapsone
Fungal	Nystatin S&S or fluconazole ^c	Fluconazole recommended in high-risk recipients (e.g., simultaneous pancreas-kidney or liver-kidney transplant recipients, history of coccidioidomycosis, or patients who live in endemic areas)
CMV	Acyclovir, valganciclovir, ganciclovir (for CMV prophylaxis protocol, see Box 110.2)	Acyclovir (or famciclovir or valacyclovir) for HSV and VZV prophylaxis for patients not on CMV prophylaxis

^aDuration of therapy may vary among centers. Restart TMP-SMX prophylaxis for 3–6 months after any corticosteroid pulse or antibody treatment.

^bListed in order of preference. Dapsone is preferred over atovaquone for patients without anemia or G6PD deficiency. Consider adding fluoroquinolones or cephalexin (or other agents) for antibacterial activity for higher risk patients.

^cWe advocate lifelong therapy in patients with history of coccidioidomycosis or in those who live in endemic areas.

CMV, Cytomegalovirus; G6PD, glucose-6-phosphate dehydrogenase; HSV, herpes simplex virus; TMP-SMX, trimethoprim-sulfamethoxazole; UTI, urinary tract infection; VZV, varicella zoster virus.

MANAGEMENT AND PROPHYLACTIC THERAPY FOR SELECTED INFECTIONS

Suggested prophylactic therapy in kidney transplant recipients is shown in Table 110.3.

Cytomegalovirus Infection

CMV infection may cause primary infection in a seronegative recipient (donor seropositive, recipient seronegative), reactivation of endogenous latent virus (donor seropositive or seronegative, recipient seropositive) or superinfection with a new virus in a seropositive recipient (donor seropositive, recipient seropositive). Primary CMV infection is usually more severe than reactivated infection or superinfection.

Clinical Manifestations

CMV infection may be asymptomatic, manifesting as a mononucleosis-like syndrome, an influenza-like illness with fever and leukopenia or thrombocytopenia, or a severe systemic or tissue-invasive CMV disease. Gastroenteritis is the most common tissue-invasive CMV disease observed in solid organ transplantation but any organ system may be affected. Patients may present with esophagitis, gastroenteritis, hepatitis, pneumonitis, carditis, encephalitis, chorioretinitis, and even otitis. The spectrum of CMV-induced kidney injury includes tubulointerstitial nephritis, CMV glomerulopathy, and rarely CMV-induced thrombotic microangiopathy. CMV often invades the transplanted allograft due to aberrant immune response within the allograft. Without prevention strategies (discussed in a later section), CMV infection typically occurs during the first 3 months after transplant. Delayed-onset CMV infection refers to the occurrence of CMV infection 3 to 6 months after completion of antiviral prophylaxis, whereas late-onset CMV infection refers to the development of CMV infection 1 year after transplantation. Suggested risk factors for delayed-onset CMV infection include D+/R– serostatus, allograft rejection, overimmunosuppression, severe lymphopenia, and lack of CMV-specific immunity.

Immunomodulating Effects of Cytomegalovirus Infection

CMV infection is associated with immune modulation and dysregulation of helper and suppressor T cells. It may be a risk factor for acute and chronic allograft rejection, secondary infection with opportunistic agents (e.g., *Pneumocystis*, *Candida*, *Aspergillus*), and reactivation of human herpesvirus 6 (HHV-6) and HHV-7 and may favor development of PTLD. CMV infection may also accelerate HCV infection and

the development of posttransplantation diabetes mellitus (also known as *new-onset diabetes after transplantation*; see Chapter 111).

A Swiss transplant cohort study involving heart, kidney, liver, and lung transplant recipients demonstrated significantly increased risk of graft rejection within 4 weeks after detection of CMV replication with the highest risk seen in lung transplants and the lowest risk in kidney grafts. There was no significant difference in rejection risk among patients with asymptomatic untreated, asymptomatic treated, and symptomatic CMV replication.¹⁰ In recipients of liver and lung transplants, donor or recipient CMV seropositivity was associated with increased incidence of rejection even in the absence of detectable CMV DNA or disease. Although the cause-and-effect relationship of CMV infection and allograft rejection remains hypothetical, the common mediator may be the release of inflammatory cytokines. It also remains possible that immunosuppression reduction in the context of CMV DNAemia may increase the risk of acute rejection, whereas intensification of immunosuppression to treat rejection may trigger CMV reactivation.

Risk Factors for Cytomegalovirus Infection

High-risk donor and recipient serostatus (donor CMV seropositive, recipient CMV seronegative) and the use of blood products from CMV-seropositive donors are well-established risk factors for CMV infection. Other risk factors include antilymphocyte antibodies, prolonged or repeated courses of antilymphocyte preparations, high doses of maintenance immunosuppression, acute rejection episodes, comorbid illnesses, concomitant HHV-6 and HHV-7 viral infections, neutropenia, hypogammaglobulinemia, and lack of CMV-specific CD4⁺ and CD8⁺ T cells. Belatacept-based maintenance immunosuppression is associated with an increased risk of CMV primary infection and a prolonged course of viral replication in CMV high-risk patients.¹¹ CMV infection is lower among kidney transplant recipients receiving mTOR inhibitor-based compared with calcineurin inhibitor (CNI)-based regimens.¹²

CMV Screening and Prevention

Surveillance for CMV is best performed using PCR-based methods for CMV DNA. Whole blood may result in higher CMV DNA results than plasma, hence it is best to use one specimen type for comparison. CMV DNA assay is highly specific and sensitive for the detection of CMV DNAemia. Antigen-based methods (e.g., testing for pp65) have largely been replaced by CMV DNA testing, and the former may also

be falsely negative in the setting of leukopenia. Quantitative CMV assays in patients with invasive colitis and gastritis or neurologic disease, including chorioretinitis, are often negative and diagnosis may require invasive testing and biopsies. CMV DNAemia is rarely useful as a predictor of CMV retinitis and diagnosis is generally based on ophthalmologic examination. The detection of CMV DNA in vitreous fluid may be helpful in guiding the diagnosis of CMV retinitis.¹³

CMV prevention can be achieved by prophylaxis or preemptive therapy. Prophylaxis involves antiviral therapy beginning in the immediate postoperative period and continuing for 3 to 6 months. Preemptive therapy entails close CMV surveillance after transplant and initiation of treatment in patients who are found to reach a certain threshold of positive CMV DNA by PCR (threshold to treat may differ among centers). Oral acyclovir (or famciclovir or valacyclovir) is necessary for HSV and VZV prophylaxis in patients not receiving anti-CMV prophylaxis. A preemptive approach (with once weekly surveillance for 3–4 months) and universal prophylaxis appear to be comparably effective for preventing CMV disease among D+/R– and/or R+ kidney transplant recipients, and both are recommended in the 2018 International Consensus Guidelines. However, prophylaxis may be preferred in donor and/or recipient seropositive patients whose risk for CMV may be increased, and a longer duration of prophylaxis (i.e., 6 months) may also be more effective.¹³ CMV surveillance weekly for 8 to 12 weeks after the end of prophylaxis may be considered in patients at increased risk for postprophylaxis CMV disease. Retrospective studies demonstrated that secondary prophylaxis after successful treatment completion is not protective against CMV relapse.^{13,14} Suggested CMV prophylaxis protocol is shown in [Box 110.2](#).

Treatment

The Third International Consensus Guidelines and the AST Infectious Diseases Community of Practice Guidelines on the Management of CMV in Solid-Organ Transplantation are summarized in [Box 110.3](#). Mutation in the viral kinase *UL97* gene that prevents phosphorylation of ganciclovir to its active form confers ganciclovir resistance. Cidofovir and foscarnet should be reserved for those with ganciclovir-resistant strains because of their associated nephrotoxicity and potential synergistic nephrotoxicity with CNIs. Mutation in the *UL54* gene, which encodes for CMV DNA polymerase (a target for all currently available systemic anti-CMV drugs) could lead to cidofovir or foscarnet resistance or cross-resistance among ganciclovir, foscarnet, and cidofovir. Concomitant *UL97* and *UL54* mutations often confer high-level ganciclovir resistance. Letermovir is a novel anti-CMV agent with a mechanism of action distinct from that of the CMV DNA polymerase inhibitors. Letermovir inhibits CMV viral terminase complex (encoded by *UL56*, *UL51*, and *UL89* genes) and is used in CMV prophylaxis in allogeneic hematopoietic stem cell transplant recipients.¹⁵ However, letermovir has no activity against other herpesviruses, and its use for CMV prophylaxis should be complemented by acyclovir (or famciclovir or valacyclovir) for the prevention of HSV and VZV. Unlike cidofovir and foscarnet, it does not share cross-resistance with ganciclovir. Whether letermovir is effective in treating refractory or resistant CMV infection remains to be determined,¹⁶ and whether it is superior to valganciclovir in preventing CMV disease in high-risk (CMV D+/R–) adult kidney transplant recipients is being evaluated in a phase III clinical trial (study results not yet available). Maribavir is another new antiviral medication with a novel mechanism of action that has proven activity against resistant and refractory disease.¹⁷

Candida Infections

Candida infections are common in transplant recipients; *Candida albicans* and *Candida tropicalis* account for 90% of the infections. Risk factors include diabetes, high-dose corticosteroids, broad-spectrum antibacterial therapy, indwelling urinary tract device, and, rarely,

BOX 110.2 Suggested CMV Prophylaxis and Preemptive Protocol^a

2018 International Consensus Guidelines

D+/R–

- Antibody induction with lymphocyte depleting agents: valganciclovir 900 mg PO once a day × 6 mo
- If no antibody induction: valganciclovir 900 mg PO once a day × 3 mo^b
- After the end of prophylaxis, consider CMV DNA weekly × 3 mo^c

R+ (D– or D+)

- Valganciclovir 900 mg PO once a day × 3–6 mo
- Lung transplantation, composite tissue: valganciclovir 900 mg PO once a day × 6 mo

D–/R–

- Acyclovir^d 400 mg twice a day (for herpes prophylaxis) × 3 mo
- CMV DNA screening when clinically indicated (or if exposure to high volume of blood products)

2019 AST Infectious Diseases Community of Practice Guidelines

D+/R–

- Prophylaxis: valganciclovir (preferred), IV ganciclovir,^a or valacyclovir^a × 6 mo
- Preemptive therapy^e: weekly CMV DNA (or pp65 antigenemia) for 12 weeks after transplant. If a positive CMV threshold is reached, treat with (a) valganciclovir 900 mg orally twice a day (preferred) or (b) ganciclovir 5 mg/kg IV q12h until negative test

R+ (D– or D+)

- Prophylaxis: valganciclovir (preferred), IV ganciclovir, or valacyclovir × 3 mo
- Preemptive therapy^e: weekly CMV DNA (or pp65 antigenemia) for 12 weeks after transplant. If a positive CMV threshold is reached, treat with (a) valganciclovir 900 mg PO twice a day (preferred) or (b) ganciclovir 5 mg/kg IV q12h until negative test

^aValganciclovir, ganciclovir, acyclovir, valacyclovir, famciclovir: dose adjustment for renal function necessary.

^bAt the authors' institution, 6 months of valganciclovir prophylaxis is recommended for high risk (D+/R–) transplant recipients regardless of induction therapy.

^cAt the authors' institution, because of the increased incidence of delayed CMV DNAemia in high-risk (D+/R–) transplant recipients, preemptive CMV surveillance is routinely performed after completion of prophylaxis therapy as follows: following discontinuation of CMV prophylaxis, CMV DNA is monitored biweekly for 2 months. If after 2 months postmedication discontinuation, the patient is found to be CMV negative, CMV DNA testing is performed monthly until 1 year posttransplant.

^dMay use acyclovir, famciclovir, or valacyclovir (center dependent).

^eIf logistic support is available.

CMV, Cytomegalovirus; D+, donor CMV seropositive; D–, donor CMV seronegative; IV, intravenous; PO, orally; R+, recipient CMV seropositive; R–, recipient CMV seronegative.

donor-derived candidiasis. Fluconazole is commonly considered the drug of choice for the prevention or treatment of donor-derived candidiasis. Other azoles (such as voriconazole or itraconazole) and the echinocandins do not effectively enter the urine as active drugs and should not be used.

Superficial candidal infections involving the mouth or intertriginous areas can be treated with nystatin and topical clotrimazole, whereas candidal UTIs require systemic antifungal therapy (see [Chapter 55](#)). Whenever possible, bladder catheters, surgical drains, and urinary stents should be removed. Systemic antifungal therapy is

BOX 110.3 CMV Treatment Protocol**CMV Treatment****First-Line Treatment**

- Mild to moderate disease: valganciclovir 900 mg PO twice daily (adjusted for GFR)
- Severe or life-threatening disease or GI malabsorption: ganciclovir 5 mg/kg IV q12h (adjusted for GFR)

Second-Line Treatment (for Patients Resistant to Valganciclovir or Ganciclovir)

- Foscarnet

Comments**Monitoring During Treatment**

- Weekly plasma CMV DNA.
- Due to individual variation in virologic response, change in antiviral agent is not recommended in clinically improving patients with unchanged or rising DNAemia during the first few weeks of therapy (strong, moderate).
- Antiviral drug doses should be reduced only for worsening estimated GFR because suboptimal dosing may increase risk of clinical failure and/or resistance.
- In the setting of leukopenia, treatment with GCSF and/or withholding of myelosuppressive agents should be considered before changing valganciclovir or ganciclovir to another agent (strong, low).

Treatment Duration

- Treatment should be continued for a minimum of 2 weeks, until clinical resolution of disease and eradication of CMV DNAemia below a specific threshold (e.g., LLOQ < 200 U/mL) on 1 or 2 consecutive weekly samples.

Other

- In patients without concomitant rejection immunosuppression, reduction may be considered in the settings of severe CMV disease, inadequate clinical response, high viral loads, and cytopenia.
- Adjunctive immunoglobulin therapy is not routinely recommended.
- Secondary prophylaxis is not routinely recommended.
- In patients with prior cumulative exposure to valganciclovir or ganciclovir that exceeds 6 weeks, drug resistance should be suspected in the setting of clinical treatment failure despite 2 weeks of antiviral treatment or development of DNAemia during prophylaxis.

CMV, Cytomegalovirus; GCSF, granulocyte colony-stimulating factor; GFR, glomerular filtration rate; GI, gastrointestinal; IV, intravenously; LLOQ, lower limits of quantification; PO, orally.

indicated in the presence of any blood culture positive for *Candida* spp. (see Chapter 55).

BK Polyomavirus Infection

The clinical spectrum of BKPyV infections includes BK viruria, BK DNAemia (formerly known as BK viremia), BKVAN, and, less commonly, ureteral stenosis. The highest prevalence of BK viruria and BK DNAemia occurs at 2 to 3 months, and 3 to 6 months, respectively. BKPyV is an important cause of allograft dysfunction and graft loss and commonly manifests with an asymptomatic rise in serum creatinine during the first posttransplantation year. However, BKVAN may occur as early as the first week and as late as 6 years after transplantation. Urinalysis is usually unremarkable but may reveal pyuria, hematuria, and/or cellular casts consisting of kidney tubular cells and inflammatory cells. The American Society of Transplantation Infectious Diseases Community of Practice expert panel defines sustained BK DNAemia as more than 1000 copies/mL in two measurements within 3 weeks, or increasing to more than 10,000 copies/mL in at least 1 of 2

measurements as probable and presumptive BK nephropathy (BKN), respectively. Since PCR assays are not standardized, the sensitivity and specificity of the assays vary. A definitive diagnosis of BKVAN requires an allograft biopsy demonstrating BK viral inclusions in kidney tubular cell nuclei and occasionally in glomerular parietal epithelium (Fig. 110.1). However, a false negative BKVAN biopsy rate of 10% to 30% can occur if the biopsy sample lacks tissue from the kidney medulla where a significant reservoir of BKPyV can be found. Interstitial mononuclear inflammation, often with many plasma cells, degenerative changes in tubules, and focal tubulitis, may mimic acute rejection. BKPyV infection and acute rejection may occur simultaneously, and distinguishing between BKVAN and acute rejection or the presence of both can be a diagnostic challenge. The 2017 Banff Working Group on BKN proposed using interstitial fibrosis and intrarenal BK load levels to diagnose “definitive BKN.” BKVAN can be further categorized into 3 classes based on the percentage of tubules/ducts with evidence of viral replication, as well as Banff scores of interstitial inflammation and tubulitis. Class 1 denotes early stage BKN with favorable outcome. Class 2 and class 3 have a more pronounced negative effect on graft function, with graft failure rate reaching 50% in class 3.

Treatment strategies for BK DNAemia with or without BKVAN vary, but immunologic containment of BKV replication is universally accepted as the mainstay of therapy.¹⁸ Optimal reduction of immunosuppressive therapy may allow sufficient reconstitution of BKPyV-specific T cells to control BKPyV replication while maintaining adequate immunosuppression to prevent allograft rejection. The 2019 AST-IDCOP guidelines recommend testing all kidney transplant recipients for BK-DNAemia monthly until month 9, and then every 3 months until 2 years posttransplant. Extended surveillance after 2 years may be considered in pediatric kidney transplant recipients. Stepwise immunosuppression reduction is recommended for patients with plasma BK DNAemia more than 1000 copies/mL sustained for 3 weeks or increasing to more than 10,000 copies/mL, or in those with biopsy-proven BK-associated nephropathy. Allograft biopsy is not required for treating BK DNAemic patients whose GFR is at or close to their baseline.¹⁹

Experimental studies demonstrated that sirolimus impairs BKPyV replication in kidney tubular epithelial cells during viral early gene expression but not during viral late gene expression by interfering with mammalian target of rapamycin (mTOR)-SP6-kinase activation. Cyclosporine was similarly shown to inhibit BKPyV replication, whereas tacrolimus activates BKPyV replication and reverses sirolimus inhibition, leading to increased BKPyV replication.²⁰ In a small trial of 40 kidney transplant recipients, mycophenolate mofetil (MMF) to everolimus conversion versus 50% MMF dose reduction showed a nonsignificant trend toward improved viral clearance with everolimus conversion at 3 months follow-up.²¹ There have been no trials evaluating the safety and efficacy of leflunomide or cidofovir in BK DNAemia or BKVAN. A systematic review of 40 studies showed no graft survival benefit of adding cidofovir or leflunomide to immunosuppressive reduction for the management of BKN.²² Despite virological rationales, no trial data support switching from tacrolimus to cyclosporine-A, from mycophenolate to mTOR inhibitors or leflunomide or adding adjuncts such as intravenous immunoglobulins, cidofovir, or leflunomide (AST-IDCOP guidelines).¹⁹ Although quinolones have been shown to inhibit DNA topoisomerase and SV40 larger tumor antigen helicase, a prospective, double-blind, randomized, placebo-controlled trial of ciprofloxacin versus placebo for the prevention of BK viremia in kidney transplant recipients demonstrated no benefit of ciprofloxacin at preventing BK viremia.²³ Fluoroquinolones are not recommended for prophylaxis or therapy. BK virus-specific T-cell adoptive transfer therapy is a subject of ongoing clinical research.

Reduction in immunosuppression is not universally effective. Nonetheless, routine BK screening and early reduction in

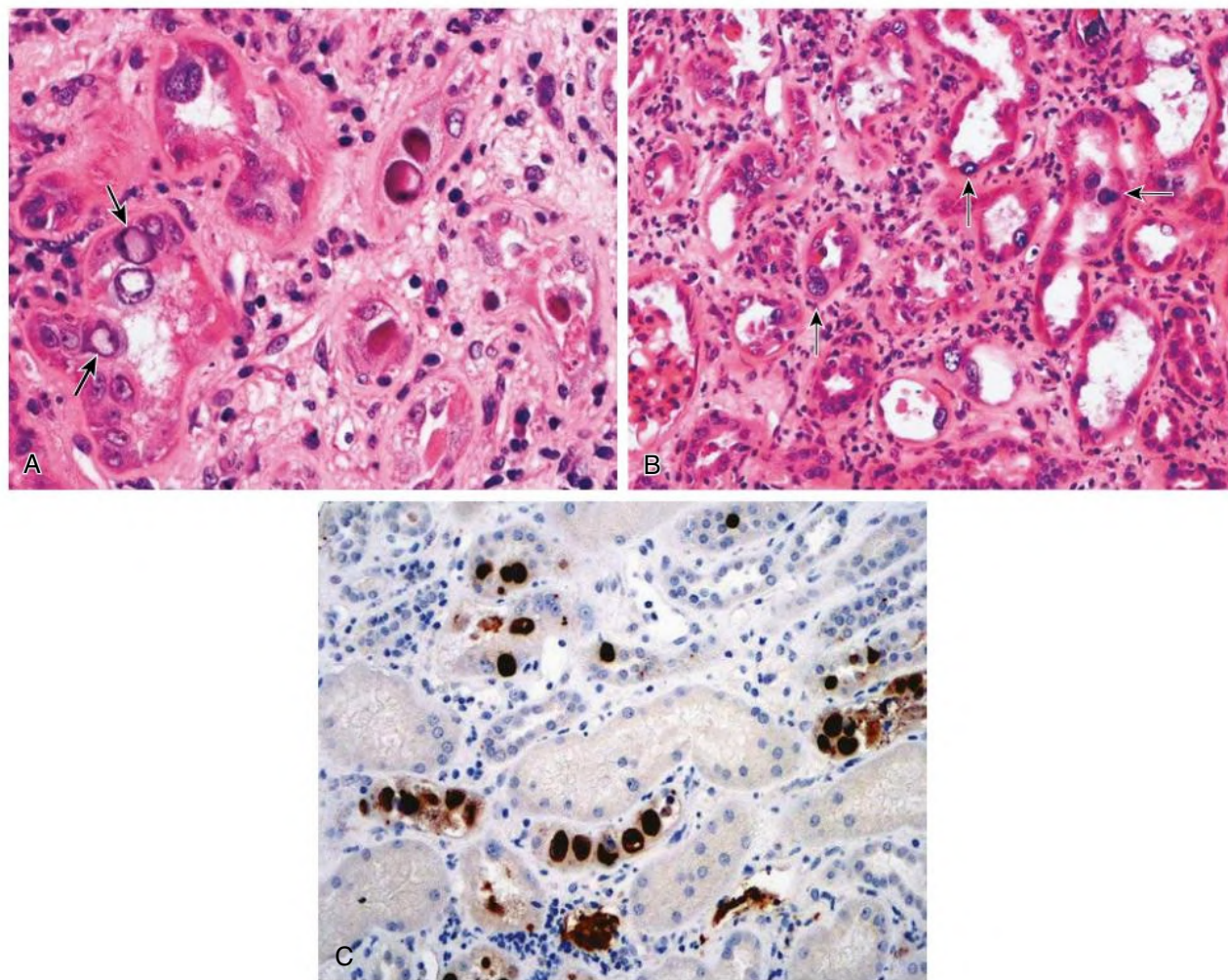


Fig. 110.1 BK Virus Nephropathy. (A) Prominent intranuclear viral inclusions are present within tubular epithelial cells (arrows). (H&E stain; original magnification $\times 400$.) (B) Tubulointerstitial nephritis with diffuse intranuclear polyomavirus inclusions (arrows). (H&E stain, original magnification $\times 200$.) (C) Immunohistochemistry staining highlights in trinuclear polyomavirus inclusions. (SV40 immunoperoxidase stain; original magnification $\times 200$.) (Courtesy Charles Lassman and William Dean Wallace, David Geffen School of Medicine at UCLA.)

immunosuppression if positive may prevent the development of BKVAN. Monitoring plasma BK alone (vs. urine BK alone or simultaneous urine and plasma BK) appears to be cost-effective because BKVAN is unusual in the absence of BK viremia. Suggested guidelines for post-transplant screening and monitoring for BKPyV replication are shown in Fig. 110.2. Although immunosuppression reduction may improve BK DNAemia and subsequent development of BKN, it may increase the risk of de novo donor specific antibody (DSA) development, underscoring the need for routine DSA surveillance. Treatment should be individualized based on each patient's immunologic risk. Although evidence is lacking, mycophenolic acid (MPA) to mTOR inhibitor switch may be considered in patients with steadily rising BK DNAemia despite MPA dose reduction or withdrawal or in those with impaired graft function to avoid the long-term nephrotoxic effect of CNI toxicity. One large international multicenter, open-label clinical trial of kidney transplant recipients randomized to receive everolimus with reduced-exposure CNI (everolimus arm) or MPA with standard-exposure CNI (MPA arm), reported less BKPyV infection (in the everolimus compared with the MPA-treated arm). Histologic evidence of organ involvement occurred in 1.2% and 2.1% of patients, respectively.²⁴

There is no consensus on the management of biopsy-proven acute rejection and concomitant BKVAN. We treat acute rejection with

antirejection therapy with subsequent reduction in maintenance immunosuppression. Intravenous immunoglobulin (IVIg) therapy may be considered in patients with persistent BK DNAemia and DSAs (which may be seen arising in association with BK viremia), in those with BKVAN and histopathologic changes that are indistinguishable from those of rejection, and in those with hypogammaglobulinemia. IVIg may be beneficial due to its direct BKPyV neutralizing activity and immunomodulatory effects. Retransplantation after allograft loss due to BKPy nephropathy can be successful if BK DNAemia is definitively cleared, independent of failed allograft nephrectomy.¹⁹

Tuberculosis

Tuberculosis (TB) is an important cause of morbidity and mortality in solid organ transplantation, with an estimated mortality rate of 20% to 30%. Over two-thirds of reported cases of active TB in solid organ transplant recipients occur in the first posttransplant year (median, 6–11 months). Although TB commonly results from inhalation of airborne bacilli, it more commonly emerges from the reactivation of dormant lesions due to immunosuppressive therapy. Hence, all potential kidney transplant candidates should undergo a purified protein derivative (PPD) skin test or preferably the interferon- γ (IFN- γ) release assays before transplantation to diagnose latent

Management of BK Viral Infection in the Kidney Transplant Recipient

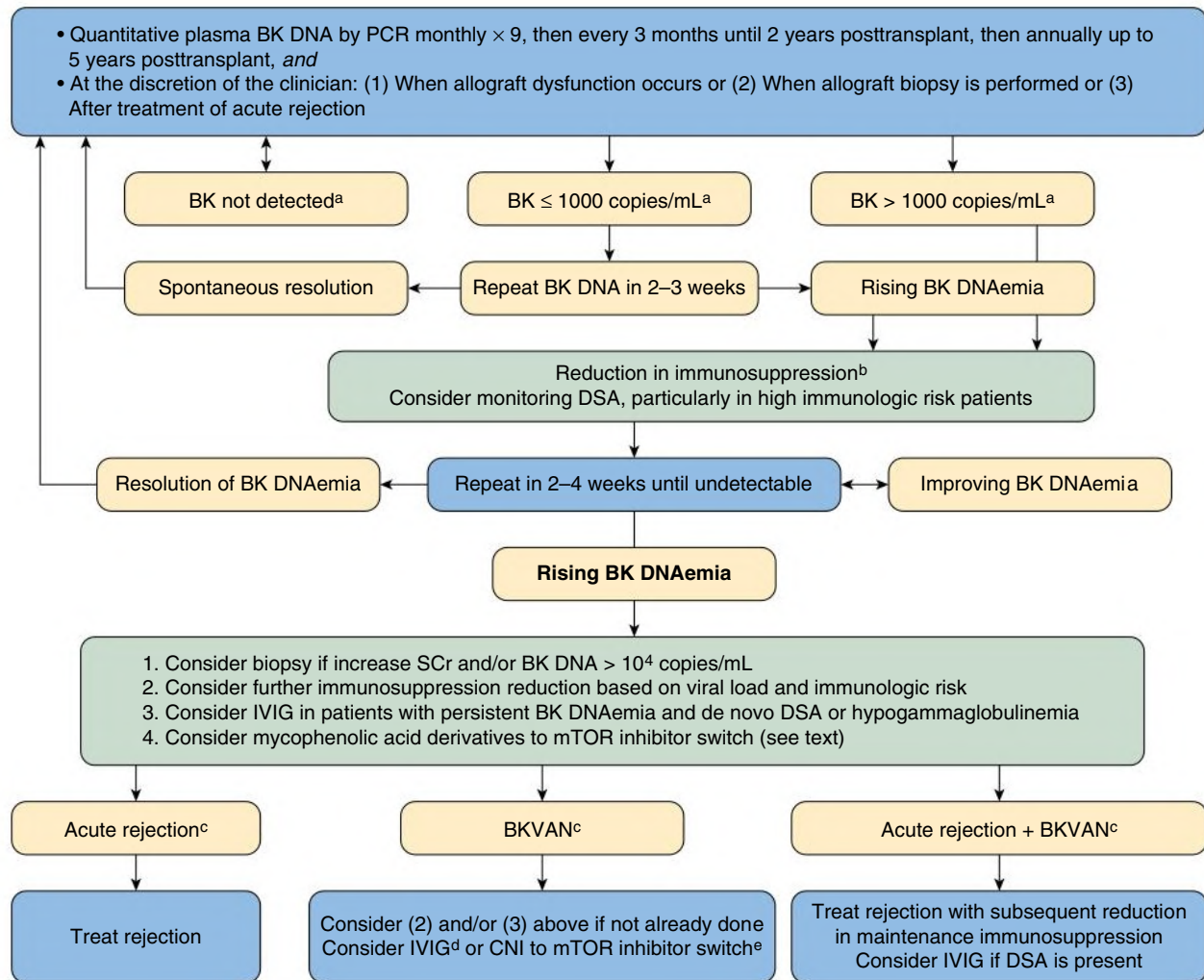


Fig. 110.2 Management of BK Viral Infection in the Kidney Transplant Recipient. ^aNo standardized PCR assays for BK virus are currently available. Cutoff levels for viral detection should be based on PCR assays used at individual institutions. Viral load of 500 copies/mL is the lowest detection level at the author's institution. ^bCommon practice: (1) Decrease or hold MFA derivatives (or antimetabolite). (2) Decrease (MFA + CNI) by 25% to 50%. (3) Decrease CNI. Less common practice: (1) Decrease prednisone with or without a decrease in CNI. (2) CNI to mTOR inhibitor switch with or without a decrease in MFA. (3) Tacrolimus to cyclosporine switch. ^cBiopsy proven. ^dIf DSA is present or histopathologic findings are indistinguishable from those of rejection. ^eMay avoid long-term nephrotoxic effect of CNI therapy. Not recommended in patients with baseline significant proteinuria (arbitrarily defined as >500 mg/24 h or at the discretion of the clinician). BKVAN, BKPyV-associated nephropathy; DSA, donor specific antibody; IVIG, intravenous immunoglobulin; MFA, mycophenolic acid; mTOR, mammalian target of rapamycin; PCR, polymerase chain reaction; SCr, serum creatinine.

TB. A positive skin test or IFN- γ release assay result or a history of TB mandates ruling out active disease. Clinical, radiologic, or culture evidence of active TB infection is a contraindication to transplantation. Isoniazid (INH) prophylactic therapy should be considered in the following situations:

- Transplant candidates with a history of positive tuberculin skin test or IFN- γ release assay result
- Patients with evidence of old granulomatous disease on chest radiograph or computed tomographic imaging and epidemiologic risk factors for TB
- Those with known prolonged exposure to a person with active TB
- Kidney transplant recipients in endemic areas
- Donor with positive tuberculin skin test or IFN- γ release assay result, or history of untreated TB
- In highly endemic areas where TB transmission is common, universal INH prophylaxis for the first year posttransplant during the period of maximum immunosuppression is recommended. In countries where there is an increased prevalence of INH-resistance strain, such as India, close monitoring and treatment of active infection is recommended over initiation of INH prophylaxis.
- In patients with a known history of adequately treated TB infection, we suggest secondary INH prophylaxis for the first 9 months after transplantation in patients with a history of extensive disease such as positive blood cultures, visceral involvement, or recent treatment

history (arbitrarily defined as <10 years before transplant). Others, however, have suggested that INH prophylaxis is not indicated for patients whose TB had been properly treated.

Strongyloides

Strongyloides is a rare but important cause of infection in transplant patients, particularly those from endemic areas such as Southeast Asia. In the presence of immunosuppression, a hyperinfection syndrome may be observed with parasitic pneumonia and gastrointestinal (GI) involvement.

EMERGING INFECTIOUS DISEASE

COVID-19

The coronavirus disease 2019 (COVID-19) pandemic caused by SARS-CoV-2 has severely impacted both immunocompetent and immunocompromised populations worldwide. Solid organ transplant (SOT) recipients, including kidney transplant recipients, are at increased risk because of their immunocompromised state.²⁵ Similar to the general population, the clinical spectrum of COVID-19 in SOT recipients spans from asymptomatic/pauci-symptomatic to acute respiratory distress syndrome with multiorgan failure and death. Common presenting symptoms include fever, dry cough, dyspnea, and fatigue. Others include diarrhea, anorexia, nausea, headache, myalgias, and anosmia. Limited retrospective studies suggest that the Omicron variant causes less severe illness compared with ancestral SARS-CoV-2 in both the general population and in SOT recipients. In addition, although vaccination response as measured by humoral or T-cell response is impaired in SOT recipients compared with those on dialysis or healthy controls,²⁶ one large single-center registry study ($n = 2151$ SOT recipients) demonstrated decreased hospitalization and mortality rates among vaccinated compared with unvaccinated SOT recipients with COVID-19 infection.²⁷

Management of Immunosuppression During COVID-19 Infection

The optimal approach to the management of immunosuppression in kidney transplant recipients infected by COVID-19 has not been well defined. Nonetheless, many transplant centers favor reduction or discontinuation of antimetabolites in symptomatic patients, particularly those with lymphopenia. Additionally, the use of lymphocyte-depleting agents such as antithymocyte globulin is generally avoided. Lymphopenia at the time of admission correlates with worse outcomes, including increased mortality and intensive care unit admission.²⁸ Rituximab causes B-cell depletion and has been shown to compromise antiviral immunity and impair COVID-19 vaccine efficacy.

A meta-analysis of randomized controlled trials conducted in the general population demonstrated that in critically ill COVID-19 patients, administration of systemic corticosteroids (dexamethasone, hydrocortisone, or methylprednisolone) was associated with lower 28-day all-cause mortality, compared with standard care or placebo.²⁹ The beneficial effect of high-dose corticosteroids may be due to their immunomodulatory and antiinflammatory effects. The Centers for Disease Control and Prevention guidelines recommend that hospitalized patients, including kidney transplant recipients requiring supplemental oxygen, receive a short course of dexamethasone (6 mg daily for 5 to 10 days) (<https://www.covid19treatmentguidelines.nih.gov/tables/therapeutic-management-of-hospitalized-adults/>). Low-dose chronic maintenance prednisone therapy is withheld until after completion of dexamethasone.

Interleukin-6 (IL-6), which stimulates the inflammatory pathways in response to infection, is commonly found to be elevated in subjects with severe COVID-19 infection,³⁰ suggesting that IL-6 could be a potential therapeutic target for patients presenting with severe

cytokine storm syndrome. Calcineurin inhibitors (CNIs) inhibit IL-6 and IL-1 pathways, and their use may confer theoretical benefits in the settings of COVID-19 infection. Hence CNIs can generally be continued without dose reduction.

Large randomized controlled trials demonstrated that adding an immunomodulator such as an IL-6 inhibitor (e.g., tocilizumab or sarilumab) or Janus kinase inhibitor (e.g., baricitinib, tofacitinib) to dexamethasone improved survival in symptomatic hospitalized patients with evidence of systemic inflammation. A phase III randomized, double-blind, parallel-group, placebo-controlled trial demonstrated that baricitinib reduced 28-day and 60-day all-cause mortality when used in conjunction with standard care, including systemic corticosteroid (e.g., dexamethasone), antiviral (remdesivir), or both (hazard ratio [HR], 0.57; 95% CI, 0.410.78; nominal $P = .0018$; and HR, 0.62; 95% CI, 0.470.83; $P = .0050$, respectively). There was a nonsignificant reduction in the frequency of disease progression (defined as progression to high-flow oxygen, noninvasive ventilation, mechanical ventilation, or death by day 28).³¹ Safety profile was similar among baricitinib-treated and standard of care alonetreated patients. Nonetheless, immunosuppressed organ transplant recipients must be closely monitored for secondary infections.

The beneficial effect of tocilizumab has not been consistently demonstrated in randomized controlled trials conducted earlier in the pandemic. However, a subsequent international, multifactorial, adaptive platform trial consisting of critically ill patients (defined as those requiring intensive care unit admission and receiving respiratory or cardiovascular organ support) demonstrated that both IL-6 receptor antagonists tocilizumab ($n = 353$) and sarilumab ($n = 48$) confer a survival benefit compared with controls ($n = 402$). The in-hospital mortality in the pooled IL-6 receptor antagonist groups was 27% vs. 36% in the control group.³²

Antiviral Therapy

Remdesivir is the first FDA-approved COVID-19 specific antiviral agent. It binds to the SARS-CoV-2 RNA-dependent RNA polymerase and inhibits viral replication via premature termination of RNA transcription. In vitro studies showed that remdesivir retains antiviral activity against the Omicron variant and subvariants. Potential adverse effects include increased serum transaminase levels and accumulation of the potentially nephrotoxic drug vehicle cyclodextrin in patients with kidney dysfunction. Remdesivir is indicated for both nonhospitalized and hospitalized patients who require no to minimal supplemental oxygen therapy (<https://www.covid19treatmentguidelines.nih.gov/tables/therapeutic-management-of-nonhospitalized-adults/>). Dexamethasone and remdesivir combination therapy should be considered in those requiring supplemental oxygen. Other FDA-approved antiviral agents include molnupiravir and ritonavir-boosted nirmatrelvir (Paxlovid). The latter should be avoided in tacrolimus or cyclosporine-treated patients due to significant drug-drug interactions. Ritonavir is a strong CYP3A4 inhibitor that is used as a druglevel boosting agent for nirmatrelvir to achieve its target therapeutic range. Similarly, concomitant use of ritonavir and drugs that are metabolized primarily by the CYP3A4 system, such as cyclosporine and tacrolimus, can result in significantly elevated CNI drug level and CNI toxicity. At the time of this writing, there is insufficient data to recommend or refute the use of molnupiravir in kidney transplant recipients. In one single-center prospective cohort study ($n = 16$ with SARS-CoV-2 [Omicron variant] and mild symptoms), one of nine patients treated with molnupiravir developed pneumonia requiring hospital admission, whereas none of seven patients treated with remdesivir progressed in disease severity. No drug-drug interaction with tacrolimus or adverse effect related to molnupiravir were observed.^{33 3031}

Management of COVID-19 in Kidney Transplant Recipients

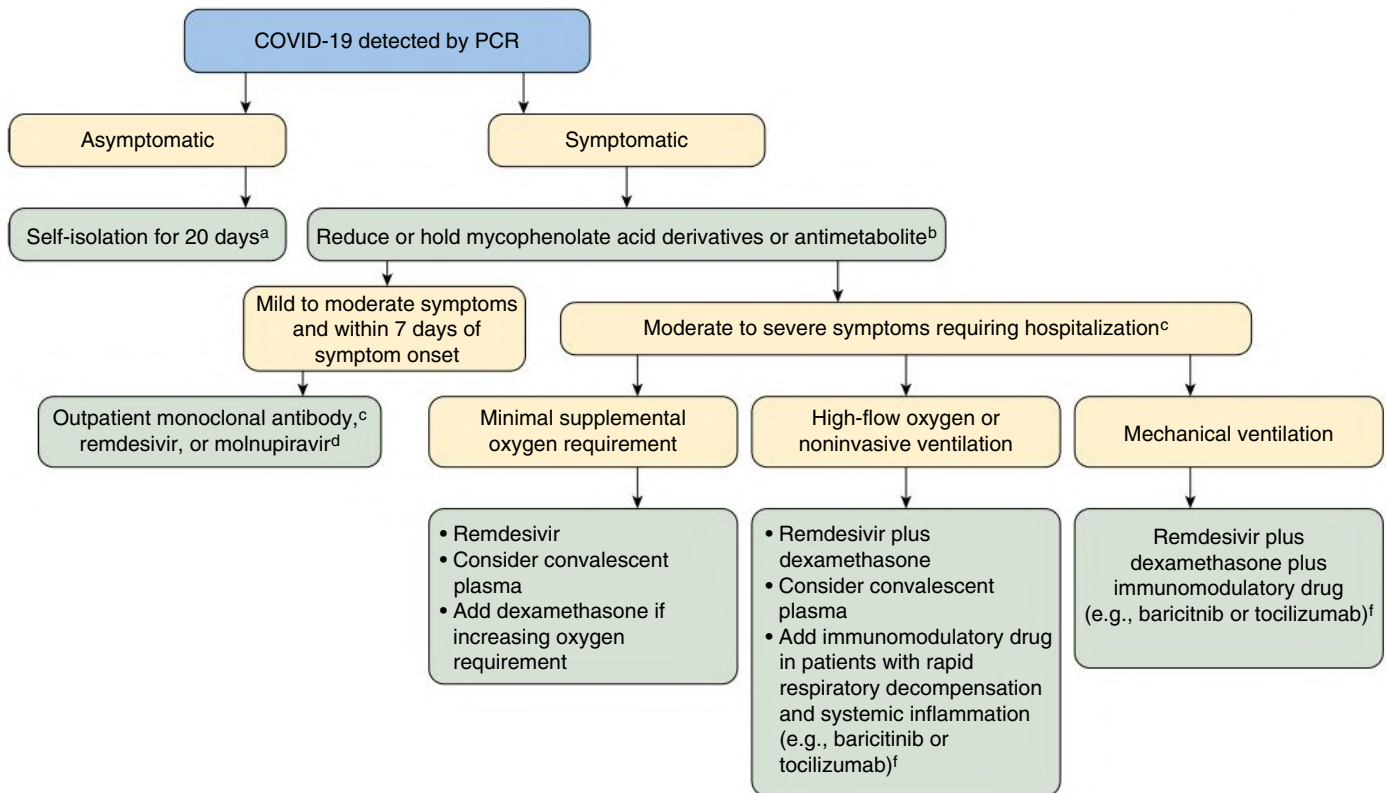


Fig. 110.3 Management of COVID-19 in Kidney Transplant Recipients. ^aSelf-isolation period may vary among geographical locations and countries. ^bConsider resuming mycophenolic acid derivatives or antimetabolites when patients are asymptomatic for 5 to 7 days (opinion based). ^cBebtelovimab retains its activity against the Omicron variant and its subvariants. ^dConsider molnupiravir for those unable to receive monoclonal antibody or remdesivir. ^eMonitor inflammatory markers at the discretion of the clinician: CBC with differentials, ferritin, D-dimer, CRP, LDH. ^fMonitor closely for secondary infections. Obtain baseline *Strongyloides* antibody, and interferon release assay (quantiferon test). Recommended prophylactic therapy: TMP-SMX daily for 3 months and valacyclovir 500 mg twice daily for 3 months. Monitor cytomegalovirus DNA at the discretion of the clinician. Low threshold for initiation of antifungal prophylaxis or treatment with prolonged steroid use. *CBC*, Complete blood cell count; *CRP*, C-reactive protein; *LDH*, lactate dehydrogenase; *MPA*, mycophenolic acid; *PCR*, polymerase chain reaction.

Monoclonal Antibody Therapy

Anti-SARS-CoV-2 monoclonal antibodies target against the SARS-CoV-2 spike protein, thereby blocking the attachment and entry of the virus into host cells. Bebtelovimab is recommended for nonhospitalized patients with mild to moderate symptoms who are at high risk of progressing to severe disease (<https://www.covid19treatmentguidelines.nih.gov/tables/therapeutic-management-of-nonhospitalized-adults/>). The United States CDC expert panel recommends against the use of bamlanivimab/etesevimab, casirivimab/imdevimab, and sotrovimab because of their markedly reduced in vitro susceptibility to the Omicron variant and its subvariants. Such variants have become the dominant SARS-CoV-2 variant in the United States. Monoclonal antibody therapy is still considered first-line therapy for high-risk outpatients with contraindication to ritonavir-boosted nirmatrelvir (Paxlovid) use due to drug-drug interactions such as the typical SOT recipient.

Anti-SARS-CoV-2 Convalescent Plasma

Plasma from donors who have recovered from COVID-19 may contain antibodies against the SARS-CoV-2 that could potentially

suppress viral replication. There is currently insufficient evidence to recommend for or against the use of convalescent plasma in kidney transplant recipients. However, given the low side effect profile of this treatment and immunologic rationale for antibody administration in immunocompromised patients, it is reasonable to consider convalescent plasma therapy in kidney transplant recipients requiring hospitalization (<https://www.covid19treatmentguidelines.nih.gov/therapies/anti-sars-cov-2-antibody-products/covid-19-convalescent-plasma/>). With the emergence of the Omicron variant, only high-titer products collected after the surge of the Omicron variant should be used. Furthermore, studies suggest that they should be given early in the course of illness for best efficacy.³⁴ A suggested algorithm for the management of COVID-19 in kidney transplant recipients is shown in Fig. 110.3 (authors' institution protocol).

GASTROINTESTINAL DISEASE

Posttransplantation GI complications are common. Selected complications are discussed.

Drug-Related Gastrointestinal Complications

MMF and to a lesser extent, enteric-coated mycophenolate sodium (EC-MPS) cause GI side effects, including nausea, vomiting, dyspepsia, anorexia, flatulence, and diarrhea. Dose reduction, temporary drug discontinuation, or administration of the drug in 3 or 4 divided doses often ameliorate the symptoms. Switching from MMF to EC-MPS can increase the maximum tolerated dose of MFA and reduce GI symptoms. MMF- and MPA-induced diarrhea may be due to direct enterocyte damage³⁵; although uncommon, Crohn's disease-like changes in the colon, erosive enterocolitis, and graft-versus-host disease-like colonic changes associated with the use of MMF have also been reported.³⁶ Proton pump inhibitor (PPI) use can reduce the dissolution of MMF by increasing gastric pH. In contrast to MMF, EC-MPS is not absorbed in the stomach, and its bioavailability is not affected by PPIs. Sirolimus, everolimus, tacrolimus, and cyclosporine may cause diarrhea in some patients. Diarrhea associated with tacrolimus use is thought to be due to increased GI motility. Sirolimus may cause oral mucocutaneous lesions that may resemble HSV or CMV infection but are culture negative. Drug-related oral ulcers usually resolve after discontinuation of the offending agent.

Infections

Posttransplantation infections of the GI tract may have a viral, fungal, or bacterial cause. Viral infections are most commonly caused by CMV and HSV; *C. albicans* and *C. tropicalis* are common opportunistic fungal infections. Leukoplakia and posttransplant lymphoproliferative disorders (PTLDs) may develop in patients with EBV infection (PTLD is discussed in a later section). Commonly encountered bacterial pathogens include *Clostridium difficile* and *Helicobacter pylori*.

Cytomegalovirus Infection

CMV can affect any segment of the GI tract. Patients may present with dysphagia, odynophagia, nausea, vomiting, gastroparesis, abdominal pain, diarrhea, or GI bleeding. Leukopenia and elevated transaminases are common. Persistent or unexplained symptoms of nausea, vomiting, or diarrhea, particularly in the early posttransplantation period or during intensification of immunosuppression, warrant further investigation with upper and lower endoscopies and biopsies. At our institution, severe CMV and candidal gastritis/esophagitis were found in a patient who was heavily immunosuppressed with antithymocyte globulin and presented with intractable vomiting 4 months after transplantation. Histologic examination of the tissue obtained from the vomitus was compatible with foveolar gastric lining detachment (Fig. 110.4).

Herpes Simplex Virus Infection

HSV infection results primarily from reactivation of endogenous latent virus, causing clinical infection within the first 1 to 2 months after transplantation. Patients commonly present with oral mucocutaneous lesions or gingivostomatitis with or without odynophagia and dysphagia. HSV esophagitis has been noted to occur in patients receiving high-dose corticosteroids and antilymphocyte preparations for acute rejection. Limited oral mucocutaneous lesions are treated with oral acyclovir; extensive infections require intravenous acyclovir or ganciclovir. For patients not receiving anti-CMV prophylaxis, acyclovir prophylaxis in the early posttransplantation period is recommended (see Box 110.2).

Fungal Infections

Candida stomatitis and esophagitis are common during the first 6 months after transplantation and are increased in patients with leukopenia, severe immunosuppression, diabetes, or concomitant infections. The use of prophylactic oral nystatin “swish and swallow” during



Fig. 110.4 Tissue Obtained From Patient's Vomitus. Histologic examination of the necrotic tissue showed numerous fungal organisms (consistent with *Candida* spp.), and degenerated squamous epithelium with detached columnar (gastric foveolar type) epithelium. Occasional cells demonstrated enlarged nuclei and intracytoplasmic granular eosinophilic inclusions, consistent with cytomegalovirus viral inclusions.

the first month after transplantation is recommended. Fluconazole prophylaxis is warranted in high-risk candidates, including liver or pancreas transplant recipients.

Helicobacter Infection

H. pylori infection is associated with a wide range of GI complications including chronic gastritis, duodenal and gastric ulcers, mucosa-associated lymphatic tissue (MALT) lymphoma, and gastric carcinoma, both in the general population and in solid organ transplant recipients. Unexplained dyspeptic or reflux symptoms should be investigated further with endoscopy and biopsy to exclude malignancy. *H. pylori*-associated MALT lymphoma in kidney transplant recipients may be less aggressive than other lymphomas, and the disorder may be cured by eradication of *H. pylori*.

Diarrhea and Colon Disorders

Diarrhea is common after solid organ transplantation because of adverse drug effects (discussed in an earlier section) and infectious pathogens. *Clostridioides difficile* (formerly known as *Clostridium difficile*) is the most commonly encountered bacterial pathogen, and CMV and norovirus are the most commonly encountered viral pathogens. Suggested treatment regimen for *C. difficile* diarrhea is shown in Table 110.4. Although norovirus often causes acute self-limited illness in immunocompetent individuals, solid organ transplant recipients can experience prolonged or intermittent diarrhea associated with chronic viral shedding. A median viral shedding period of 289 days (97–898 days) has been reported. Molecular analysis of viral strains in kidney transplant recipients with chronic intermittent diarrhea suggests that continuous genetic mutation and viral evolution may promote viral persistence.³⁷ Other less common infectious causes of posttransplantation diarrhea are shown in Table 110.4.^{35,36,38}

Diverticulitis and colonic perforations may be life-threatening and difficult to diagnose after transplantation because symptoms may be masked by immunosuppressive therapy, particularly in the early postoperative period. Diverticulitis complicated by perforation, abscess formation, phlegmon, or fistula has been reported in 1% of kidney

TABLE 110.4 Causes of Diarrhea After Solid Organ Transplantation

Pathogens	Suggested Therapies/Comments
Bacterial	
<i>Clostridioides difficile</i>	<i>C. difficile</i> ^a
<i>Campylobacter</i> spp.	• <i>First episode</i> : Oral vancomycin 125 mg 4 times daily for 10–14 days or fidaxomicin 200 mg twice daily for 10 days
<i>Salmonella</i> spp.	• <i>First relapse</i> : treatment same as first episode
<i>Shigella</i> spp.	• <i>Second relapse</i> : oral vancomycin or fidaxomicin
<i>Aeromonas</i> spp.	• <i>Third or more relapses</i> : prolonged oral vancomycin taper (125 mg 4 times a day for 10–14 days, twice times a day for a week, once a day for a week, and then every 2 or 3 days for 2–8 weeks). May consider fecal microbiota transplantation or bezlotoxumab (monoclonal antibody).
<i>Escherichia coli</i>	<i>E. coli</i>
Others: <i>Vibrio cholera</i> , <i>Yersinia enterocolitica</i> , bacterial overgrowth, chlamydia, <i>Helicobacter pylori</i>	• Enteropathogenic: fluoroquinolone, cephalosporin
	• Enterohemorrhagic: avoid antimotility agents, role of antibacterials unclear and should be avoided
Viral	
CMV	CMV (see text)
Norovirus	Norovirus
Sapovirus	• Diagnostic testing: norovirus PCR
Others: rotavirus, adenovirus, enterovirus, herpes simplex virus, SARS-CoV-2	• Treatment: supportive (volume repletion, antimotility agents), reduction of immunosuppression (inconclusive evidence), nitazoxanide (anecdotal case reports)
Parasitic	
<i>Giardia</i> , <i>Cryptosporidium</i> , <i>Cyclospora</i> , <i>Microsporidia</i> , <i>Strongyloides</i> , <i>Entamoeba</i> , <i>Isospora</i>	

^aMetronidazole is no longer considered first-line treatment because of the risk of recurrence after treatment.

CMV, Cytomegalovirus; PCR, polymerase chain reaction.

transplant recipients, and its incidence may be increased in patients with autosomal dominant polycystic kidney disease (ADPKD).³⁹

Early posttransplantation colonic perforations are largely caused by high-dose corticosteroids, diverticulitis, CMV colitis, and intestinal ischemia; perforations occurring late or years after transplantation are commonly caused by diverticulosis or malignant disease.

TRANSPLANT-ASSOCIATED MALIGNANCY

The incidence of de novo malignancies is twofold to fourfold greater in solid organ transplant recipients compared with the general population.⁴⁰ Among long-term survivors after kidney transplantation (>20 years), cancer was the most common cause of death with a functioning graft followed by CV disease.⁴¹ The intensity and duration of immunosuppression, as well as the ability of these agents to promote replication of various oncogenic viruses, are important risk factors. Table 110.5 provides a summary of the incidence of cancers related to infections in solid organ transplant recipients.⁴² The Transplant Center March Study database demonstrated a strong association of lip cancer with

TABLE 110.5 Meta-Analysis Standardized Incidence Ratios (SIRs) for Cancers Related to Infections in Transplant Recipients

Cancers	Meta-Analysis SIRs
EBV-Related Cancers	
Hodgkin lymphoma	3.89 (2.42–6.26)
Non-Hodgkin lymphoma	8.07 (6.40–10.2)
HHV-8-Related Cancer	
Kaposi sarcoma	208.0 (114–369)
HBV- and HCV-Related Cancer	
Liver	2.13 (1.16–3.91)
HPV-Related Cancers	
Cervix uteri	2.13 (1.37–3.30)
Vulva and vagina	22.8 (15.8–32.7)
Penis	5.8 (5.79–34.4)
Anus	4.85 (1.36–173)
Oral cavity and pharynx	3.23 (2.40–4.35)
Nonmelanocytic-related skin	28.6 (9.39–87.2)

EBV, Epstein-Barr virus; HBV, hepatitis B virus; HCV, hepatitis C virus; HHV, human herpes virus; HPV, human papilloma virus; SIR, standardized incidence ratio.

Modified from McCaughan JA, Courtney AE. The clinical course of kidney transplant recipients after 20 years of graft function. *Am J Transplant.* 2015;15(3):734–740.

White ethnicity and prior history of skin cancer, suggesting that ultraviolet (UV) radiation exposure is an important risk factor. A higher incidence of lip cancer was also found among transplant recipients receiving cyclosporine and azathioprine (AZA) compared with tacrolimus/MMF maintenance immunosuppression. The contributory role of cyclosporine and AZA was thought to be due to their photosensitizing or DNA damaging effects.⁴³ Other suggested risk factors for posttransplantation malignancies include older age, male sex, White ethnicity, pretransplant dialysis duration, smoking history, deceased donor organ, cumulative exposure to radiation from repeated medical imaging studies, and prior use of immunosuppressive agents.

Skin cancers are the most common de novo posttransplant tumors in the adult transplant population and occur with increasing frequency with time. Among nonskin malignant neoplasms, PTLDs are the most common type of posttransplantation malignancy. The mean time to diagnosis of different neoplasms varies with the type of organ involved. Kaposi sarcoma, PTLT, testicular cancer, cancer of the small intestine, and thyroid cancer occur early (<800 days) after transplantation. In the AZA era, nearly 50% of adult transplant recipients have one or more skin cancers by 20 years after kidney transplant. The use of MPA-containing regimen has been reported to be associated with a lower incidence of posttransplantation nonmelanoma skin cancers compared with AZA. (The effects of various immunosuppressive agents on cancer risk are discussed later.⁴⁰) Renal cell carcinoma (RCC) occurs mainly in native kidneys and is associated with acquired cystic kidney disease and dialysis duration. A bimodal distribution is observed with the highest incidence occurring in the first year and a second peak 4 to 15 years after transplant.⁴⁴ RCC occurring early after transplant is thought to reflect undetected or malignant transformation of preexisting cysts that developed during the course of end-stage kidney disease. Common cancers in the general population, including cancers of the breast, colon, prostate, lung, bladder, stomach, and pancreas, were found to occur more frequently in kidney transplant recipients in some but not all studies. The Kidney Disease Outcomes Quality Initiative guidelines suggest an individualized plan for each kidney

transplant recipient, considering the patient's medical and family history, tobacco use, and competing risk for death. We evaluate hepatitis B–positive transplant candidates and recipients for hepatocellular carcinoma with abdominal ultrasound and α -fetoprotein level every 6 months. Although practice differs among centers, screening should be considered in high-risk patients, such as those with high HBV viral load, hepatitis B e antigen–positive, hepatitis B surface antigen–positive, and Asian or African ethnicity. Screening for cervical, breast, prostate, and colon cancer should follow local guidelines for the general population. In recipients with a history of preexisting malignant neoplasms, close monitoring for recurrences is mandatory. Suggested guidelines for tumor-free waiting periods for common pretransplant malignant neoplasms are shown in [Table 110.6](#).⁴⁵

Posttransplantation Lymphoproliferative Disorder

PTLD encompasses a wide spectrum of lymphoid proliferations ranging from reactive polyclonal lesions to frank malignant monoclonal lymphomas and can be divided into four subtypes based on morphology, clonality, and molecular criteria ([Table 110.7](#)).^{45,46} Most PTLDs are non-Hodgkin lymphoma of B-cell origin, 80% to 90% of which are linked to EBV infection. However, the incidence of EBV-negative PTLD has increasingly been reported. PTLD commonly occurs in the first year after transplant, although the cumulative incidence increases with time after transplantation. Registry studies demonstrate that PTLD occurs at a median of 18 to 18.5 months after transplantation. Notably, EBV-negative PTLD generally manifests later after transplant (>5 years) and has been suggested to account for the bimodal distribution pattern of PTLD occurrence, with early cases being predominantly EBV positive and late cases being EBV negative.⁴⁰ Another registry study suggests that late-occurring PTLD (8–10 years) is less likely to be due to EBV lymphoproliferation and more likely to be a consequence of aging and duration of immunosuppression exposure.⁴⁷ The incidence of EBV-associated PTLD by PTLD subtypes and time of onset after transplant is shown in [Table 110.8](#).⁴⁵ Mortality is greater with PTLD than with lymphomas in the general population.

EBV-Positive Versus EBV-Negative PTLD

Immunosuppression-related decrease in T-cell immune surveillance may play a major contributory role in EBV-positive PTLD. In immunocompetent hosts, EBV-specific CD8⁺ effector and memory T cells are responsible for controlling EBV-infected B cells from uncontrolled proliferation and transformation. The pathogenesis of EBV-negative PTLD remains speculative. Proposed hypotheses include hit-and-run EBV infection (disappearance of EBV following an initial infection that leads to PTLD), viral infection other than EBV (e.g., cytomegalovirus), persistent antigen stimulation by the graft, and long-term immunosuppression. Molecular-genomic studies revealed that EBV-negative PTLD shares many genomic and transcriptomic features with diffuse large B-cell lymphoma in immunocompetent patients, whereas EBV-positive PTLD has fewer genomic abnormalities. Although the molecular genetic separation between EBV-positive and EBV-negative PTLD is well-defined, EBV status is not prognostic or predictive with respect to treatment response. In a subset of patients, reduction in immunosuppression alone is effective regardless of EBV status.

Risk Factors

Risk factors for PTLD can be classified into those associated with EBV or other infectious agents, the type of organ transplant, recipient age, the type and intensity of immunosuppression, and human leukocyte antigen (HLA) mismatch among others ([Table 110.9](#)). Selected risk factors are discussed.

Epstein-Barr virus infection. During primary infection, EBV is incorporated into B lymphocytes and establishes lifelong latency.

Immunocompetent hosts mount an antibody response and, more importantly, an EBV-specific cytotoxic T-cell immune response. EBV-naïve transplant recipients lack both virus-specific antibodies and EBV-specific T cells; therefore an EBV-negative recipient of an EBV-positive donor is at greatest risk for developing PTLD.

Type of immunosuppression

Calcineurin inhibitors. Cyclosporine and tacrolimus may enhance the development of EBV-associated PTLD by directly promoting survival of EBV-infected B cells, presumably via the inhibition of EBV-transformed cells from apoptosis. Although single-center retrospective studies demonstrated that tacrolimus increased the risk for PTLD twofold to fivefold compared with cyclosporine, tacrolimus use has not been consistently shown to be associated with increased PTLD risk compared with cyclosporine.⁴⁸

mTOR inhibitors. Sirolimus has a strong antiproliferative effect on PTLD-derived B-cell lines. Although there are theoretical reasons to suspect that mTOR inhibitors might decrease the risk of PTLD, one retrospective study demonstrated that sirolimus-containing regimens are associated with a 1.2-fold higher incidence of PTLD compared with no sirolimus.⁴⁹ The UNOS/OPTN database similarly demonstrated that in EBV-negative kidney transplant recipients, mTOR inhibitor + tacrolimus (Tac) combination therapy was associated with a nearly twofold increased PTLD risk compared with MMF/MPS + Tac combination therapy, whereas MMF/MPS + cyclosporine A (CsA) immunosuppression was associated with lower PTLD risk.⁵⁰ In contrast, a favorable effect of mTOR inhibitor on PTLD has also been reported. In a case series of 13 kidney transplant recipients with PTLD, immunosuppression reduction (MMF or AZA dose reduction along with discontinuation of Tac or CsA) in conjunction with initiation of sirolimus alone was effective in induction of remission in 4 patients with diffuse large B-cell PTLD after a median time of 12 weeks.⁵¹ The potential beneficial effect of sirolimus on PTLD risk reduction was also shown in a retrospective study of 523 heart transplant recipients. Of these, 307 underwent CNI to sirolimus conversion therapy with complete withdrawal of CNI at a median of 1.1 years after heart transplant. Over median follow-up of 10 years posttransplantation, 0.65% of patients (2/307) in the sirolimus conversion group developed PTLD compared with 10.18% of patients (22/216) in the CNI group (adjusted HR, 0.13; $P = .009$). Of note, however, 7.9% of patients in the CNI maintenance group were recipients of combined heart-lung transplantation, whereas only 0.65% of patients in the sirolimus group received combined heart-lung transplant. Whether the overall degree of immunosuppression accounted for the higher incidence of PTLD in the CNI-maintenance group could not be determined from the study.⁵²

Antimetabolites. The use of antimetabolites such as AZA and MMF has not been consistently shown to be associated with an increased risk for PTLD.

Induction agents. Whereas induction therapy with muromonab CD3/OCT3 (removed from the US market in 2010) or antithymocyte globulin may increase PTLD risk, the use of IL-2 receptor inhibitors (basiliximab and daclizumab [removed from the market in 2018]) or anti-CD52 antibody (alemtuzumab) induction has not been reported to increase PTLD risk. Analysis of the UNOS/OPTN database demonstrated that thymoglobulin was associated with significantly increased PTLD risk, whereas alemtuzumab, basiliximab, and daclizumab were associated with a borderline protective effect ($P = .06$). In this study maintenance therapy with an mTOR inhibitor was strongly associated with PTLD, in contrast to earlier reports suggesting its beneficial effect on PTLD because of its antiproliferative and antiangiogenic effects. Although both alemtuzumab and thymoglobulin are T-cell–depletional agents, the former has been shown to have a more

TABLE 110.6 Tumor-Free Waiting Period for Common Pretransplantation Malignancies^{a,b}

No Waiting Time	
Long-standing history of MGUS; hematology-oncology consult is advisable in patients with newly diagnosed monoclonal gammopathy	
No Waiting Time if Cure at the Time of Transplantation	
Incidental renal cell carcinoma, in situ carcinoma of bladder, in situ carcinoma of cervix, basal cell carcinoma	
Waiting Time Varies With Staging, Tumor Size	
Breast cancer	2–5 years 2-year waiting time for ductal carcinoma in situ Patients with stage III or IV breast cancer should be advised against transplantation
Prostate cancer	Old paradigm: 2-year tumor-free waiting time Current trend ^b : shorter waiting time is acceptable for patients with grade group 1 or 2 prostate cancer (Gleason score ≤ 6 or Gleason 3 + 4, respectively)
Renal cell carcinoma	2 years if <5 cm 5 years if >5 cm or <5 cm with invasion
Skin SCC ^{b,c}	Old paradigm ^d : 0- to 2-year waiting time New paradigm ^d : low-risk SCC: surgical excision with clear margins or Mohs surgery and no waiting time High-risk SCC with no perineural invasion: surgical excision with clear margins or Mohs surgery and 2-year waiting time High-risk SCC with perineural invasion: surgical excision with clear margins or Mohs surgery ± adjuvant radiation therapy and waiting time of 5 years
Melanoma ^{b,c}	High-risk SCC with local nodal disease: surgical excision, lymph node dissection, adjuvant radiation therapy, and 5-year waiting time Old paradigm: 5-year waiting time New paradigm ^d : in situ: wide excision and no waiting time Stage Ia: wide local excision and 2-year waiting time Stage Ib/Ila: wide local excision and ± sentinel lymph node biopsy and 2- to 5-year waiting time Stage IIb/Ilc: wide local excision and ± sentinel lymph node biopsy and 5-year waiting time Stage III or IV: not a transplant candidate
PTLD (retransplantation)	At least 1–2 years ^b
Waiting Time 2–5 Years^b	
2-year waiting time: invasive bladder, uterine body, Wilms tumor	
2- to 5-year waiting time: lymphoma, invasive cervical carcinoma, colorectal carcinoma (at least 5 years for Dukes B1 or higher)	

^aCertain cancers may recur despite a tumor-free waiting period.

^bOncology evaluation or consultation with the Israel Penn International Transplant Tumor Registry at <https://ipittr.uc.edu> may be invaluable.

^cDermatology consultation is recommended.

^dMittal A, Colegio OR. Skin cancers in organ transplant recipients. *Am J Transplant*. 2017;17(10):2509–2530.

MGUS, Monoclonal gammopathy of undetermined significance; PTLD, posttransplantation lymphoproliferative disorder; SCC, squamous cell carcinoma.

TABLE 110.7 WHO Classification of Posttransplant Lymphoproliferative Disorder

Categories	Histopathology	EBV Association/Comments
Early lesions	- Plasmacytic hyperplasia: many polyclonal plasma cells with occasional immunoblasts and lymphocytes - Infectious mononucleosis–like: many immunoblasts, with T lymphocytes and plasma cells - Floral follicular hyperplasia (included in the 2016 WHO classification of lymphoid neoplasms)	- Usually EBV positive - No or very focal clonality, no atypical morphology
Polymorphic PTLD	- Lymphocytes, plasma cells, immunoblasts, may be Reed-Sternberg–like cells - Often monoclonal, may have necrosis	- Often EBV positive - Cellular atypia, gene rearrangements may occur
Monomorphic PTLD	- Fulfills WHO criteria for non-Hodgkin lymphoma or plasma cell neoplasm - B cell (most common) - Diffuse large B-cell lymphoma, not otherwise specified - Burkitt lymphoma - Plasmacytoma-like lesions and plasmacytoma - Other - T cell/NK cell: EBV negative - Peripheral T-cell lymphoma, not otherwise specified - Hepatosplenic lymphoma - T-cell/NK-cell lymphoma - Other	- EBV less often positive - Uniform population of neoplastic cells with clonality and gene rearrangements
Hodgkin lymphoma–like PTLD	Rare, usually mixed cellularity type	EBV positive

EBV, Epstein-Barr virus; NK, natural killer; PTLD, posttransplantation lymphoproliferative disorder; WHO, World Health Organization.

pronounced B-cell-depleting effect. We speculate that depletion induction is not an independent risk factor for PTLD, but rather that maintenance drug selection and the balance between B-cell and T-cell depletion may determine the risk of PTLD. Nonetheless, aggressive PTLD attributed to alemtuzumab induction has been reported. It is speculated that the overall intensity of immunosuppression increases PTLD risk by decreasing host cytotoxic T cells directed against grafted EBV-infected B lymphocytes.

Belatacept. Belatacept is a nonantigen-specific biologic agent that blocks T-cell costimulatory signals, hence preventing T-cell activation. Phase III clinical trials suggest that its use in transplant recipients with pretransplant EBV seronegativity is associated with an increased risk for PTLD. However, a systematic review suggested that belatacept use at different dosages and in patients who were EBV seronegative

or seropositive before transplant confers no additional risk compared with CNIs immunosuppression.⁵³ Nonetheless, the black box warning for belatacept use in EBV-seronegative transplant candidates remains, and its use should be avoided in transplant recipients with pretransplant EBV-seronegative or unknown EBV status.

Donor-recipient HLA mismatch and HLA type. Retrospective analysis of more than 172,000 kidney, heart, pancreas, and lung recipients transplanted in the United States demonstrated that compared with recipients who had 2 HLA-donor-recipient (DR) mismatches, those with zero or 1 mismatch had reduced diffuse large B-cell lymphoma (DLBCL) risk (*0 DR mismatch*: incidence rate ratio [IRR], 0.76; 95% confidence interval [CI], 0.61–0.95; *P* = .0149; *1 DR mismatch*: IRR, 0.83; 95% CI, 0.69–1.00). Analysis of individual HLA type showed that recipient HLA-B38 was associated with a 1.48-fold increased DLBCL risk, whereas recipient HLA-B58 and HLA-DR13 were associated with reduced DLBCL risk.⁵⁴

TABLE 110.8 Incidence of EBV-Associated PTLD by Subtypes and Time of Onset After Transplant

PTLD Subtypes	EBV Association	Onset After Transplant
Early lesions	Almost 100% EBV positive	Most early PTLD
Polymorphic PTLD	>90% EBV positive	Variable
Monomorphic PTLD	Both EBV positive and EBV negative	- EBV positive: most occur within the first 3 years after transplant - EBV negative: late occurring PTLD (>5 years after transplant)
Hodgkin lymphoma–like PTLD	>90%	

EBV, Epstein-Barr virus; PTLD, posttransplantation lymphoproliferative disorder.

Clinical Manifestations

PTLD may manifest with constitutional symptoms such as fevers, night sweats, malaise, and weight loss or with localized symptoms of the respiratory tract (infection or mass, including tonsillar or even gingival involvement), GI tract (diarrhea, pain, perforation, bleeding, mass), or central nervous system (CNS; headache, seizure, confusion). Other clinical manifestations may include lymphadenopathy or symptoms related to allograft dysfunction or compression of surrounding structures. Extranodal involvement occurs in more than two-thirds of patients. GI tract involvement occurs in 20% to 30% of cases, and solid graft organs in 10% to 15%. In contrast to lymphomas in the general population, the CNS is frequently involved in PTLD, occurring in up to 25% to 30% of patients, and can be the only site of disease. Early diagnosis requires a high index of suspicion and radiologic findings. Although computed tomography and magnetic resonance imaging are the most commonly used imaging modalities in PTLD, positron emission tomography-computerized tomography (PET-CT) has emerged as a potentially useful adjunct test for the diagnosis, staging, and

TABLE 110.9 Risk Factors for PTLD

Risk Factors	Comments
- Pretransplant EBV seronegative and primary EBV infection - Young recipient age (children compared with adults)	- Well-established risk factor - Higher incidence of PTLD in children compared with adults has been attributed to pretransplant EBV-naïve status in pediatric population
PTLD incidence by type of organ transplant: - Small bowel and multiviscera (up to 33%) - Lung, heart, and heart-lung (2%–10%) - Pancreas or simultaneous kidney-pancreas (2%–3%) - Liver (1%–4%) - Kidney (1%–2%)	More intensive immunosuppression and the amount of donor-derived lymphoid tissue transferred at organ transplantation may account for a high incidence of PTLD in multiorgan and bowel transplantation
Type of immunosuppression: - Belatacept (increased risk in pretransplant EBV seronegative recipients) - Lymphocyte-depleting agents (antithymocyte globulin > alemtuzumab) - CNIs - mTOR inhibitors (inconclusive or contradictory data) - Antimetabolites (inconsistent data) - Others (less well-established or controversial): HLA matching/HLA type, Black ethnicity, male sex, CMV infection/disease, HCV, pretransplant malignancy, HHV-8 and simian virus 40 infections, preexisting chronic immune stimulation, recipient MGUS	- Belatacept use is contraindicated in patients with pretransplant EBV-seronegative or unknown EBV status - Overall intensity of induction and maintenance therapy may play a more important role than any single agent - Duration of immunosuppression 2 DR mismatches compared with 0–1 DR mismatch

CMV, Cytomegalovirus; CNI, calcineurin inhibitor; EBV, Epstein-Barr virus; DR, donor-recipient; HCV, hepatitis C virus; HHV, human herpesvirus; HLA, human leukocyte antigen; MGUS, monoclonal gammopathy of undetermined significance; mTOR, mammalian target of rapamycin; PTLD, posttransplantation lymphoproliferative disorder.

assessment of PTLD treatment response.⁵⁵ Tissue sampling is necessary for a definitive diagnosis and subcategorization of PTLD.

Treatment

Immunosuppression reduction is the first-line of treatment. Although their use has not been consistently demonstrated to increase PTLD risk, azathioprine and MPA derivatives are generally discontinued because of the theoretical net protumor effect of overimmunosuppression. CNI dose (or target level) can be reduced by 50% to 75% at the discretion of the clinician. Low-dose prednisone 5 mg daily can be continued to prevent allograft rejection. The 2019 AST-IDCOP do not recommend for or against switching to an mTOR inhibitor because of insufficient data.⁵⁵ Although a direct antitumor effect of sirolimus cannot be excluded, it is speculated that the beneficial effect of conversion to sirolimus monotherapy (or sirolimus and prednisone dual therapy) may be due to a reduction in overall immunosuppression. Restaging should be performed 2 to 4 weeks after immunosuppression reduction. For critically ill patients and those with monoclonal tumors or extensive disease, immunosuppression should be drastically reduced or discontinued. Currently reducing immunosuppression and risk-stratified sequential treatment with rituximab followed by cyclophosphamide, hydroxydaunorubicin or doxorubicin, oncovin or vincristine, prednisone (R-CHOP) is considered standard of care for polymorphic and monomorphic diffuse large B-cell lymphoma-like PTLD irrespective of EBV status (rituximab at 375 mg/m² weekly for 4 weeks, followed by CHOP every 3 weeks with granulocyte-stimulating factor support). Surgical resection with or without adjunctive local radiation can be used

for localized disease. Local radiation has been advocated to treat PTLD involving the CNS, but chemotherapy is still recommended by some experts in the field. Poor prognostic factors include multiple-site versus single-site involvement, tumor monoclonality, graft organ involvement, advanced age, CD20-negative large cell lymphomas, and recipient EBV-negative serostatus. Other poor prognostic factors include World Health Organization performance status score of 3 or 4 (a score of 3 is defined as confined to bed or a chair for more than 50% of waking hours, and a score of 4 as completely disabled), late onset of disease (>1 year after transplant), elevated lactate dehydrogenase (LDH), CNS disease, severe organ dysfunction, acute kidney injury at diagnosis, impaired kidney function, and T-cell disease.⁴⁵ Suggested treatment modalities for PTLD are shown in Table 110.10. Acyclovir and ganciclovir are of unproven benefit. A systematic review of 31 studies showed no beneficial effect of antiviral prophylaxis on the incidence of PTLD in high-risk EBV-naïve transplant recipients. The findings were consistent across all types of solid organ transplants, age groups, prophylactic or preemptive therapy, duration of antiviral prophylaxis, or different antiviral agents (acyclovir, valacyclovir, ganciclovir, valganciclovir).⁵⁶ Although arginine butyrate (a thymidine kinase-inducing agent) and ganciclovir combination therapy appears to have a promising role in EBV-associated PTLD, clinical trials evaluating its safety and efficacy have not been done.

EBV Monitoring and Preemptive Treatment Strategies

Most patients with EBV+ PTLD were shown to have a significantly higher viral load than those without PTLD. Posttransplantation EBV surveillance and preemptive treatment strategies including

TABLE 110.10 PTLD Treatment Modalities

Treatment	Indications	Comments
IS reduction	Mainstay of therapy in all PTLD subtypes	Factors predictive of a poor response to IS reduction alone: elevated LDH, organ dysfunction at diagnosis, multiple organ involvement, bulky disease (tumor deposit > 7 cm), advanced stage (Ann Arbor stage III or IV), age > 50 years
Rituximab	First-line treatment for CD20+ PTLD not responsive to IS reduction alone	
Chemotherapy	- For those who do not achieve complete remission despite IS reduction and rituximab therapy - Currently, IS reduction and risk-stratified sequential treatment with rituximab followed by CHOP (aka R-CHOP) are considered standard of care for polymorphic and monomorphic diffuse large B-cell lymphoma - Other indications: aggressive disease at presentation, lesions not amenable to surgery among others	- CHOP-based therapy is the most widely used regimen - Increased infectious risk - G-CSF use may decrease morbidity and mortality
Surgery	- For localized disease in conjunction with IS reduction - Palliative care (symptomatic relief)	
Radiation therapy	- For localized disease in conjunction with surgery and IS reduction - Palliative care (symptomatic relief)	
Antiviral therapy	- Acyclovir and ganciclovir are of unproven benefit because their activity is dependent on intracellular phosphorylation by virally coded thymidine kinase - EBV-driven lymphomas do not express thymidine kinase	Some centers continue to use antiviral therapy based on limited studies suggesting that lytic viral DNA (not just latent viral DNA) is also present in established PTLD
mTOR inhibitors	Insufficient data to recommend or refute its use in the treatment of PTLD	
Adoptive immunotherapy (adoptive transfer of EBV-specific cytotoxic T cells)	Relapsed or refractory PTLD	- EBV-positive PTLD only - Expensive, not readily available, time consuming
Retransplantation	- Should be disease-free for at least 1–2 years prior to retransplantation - Consultation with hematology-oncology recommended	

CHOP, Cyclophosphamide, hydroxydaunorubicin or doxorubicin, oncovin or vincristine, prednisone; EBV, Epstein-Barr virus; G-CSF, granulocyte-colony stimulating factor; IS, immunosuppression; LDH, lactate dehydrogenase; PTLD, posttransplantation lymphoproliferative disorder.

immunosuppression reduction, rituximab therapy, and/or adoptive transfer of EBV-specific T cells may have a beneficial effect on the incidence of PTLT. Rituximab-mediated elimination of B cells may prevent transmission of EBV to the recipient, because EBV persistence requires a latent infection in recipient B cells. Single-center retrospective study in pediatric kidney transplant recipients demonstrated that a high-PCR EBV viral load (maximum peak EBV viral load above 59,909.5 copies/mL) was a significant and independent predictor of PTLT. Ten of 103 patients developed nonearly lesions PTLT, and 5 of 103 patients developed early lesions PTLT. In all cases EBV viral load was detected prior to the clinical diagnosis of PTLT.⁵⁷ Of 5 patients who received rituximab, none developed PTLT in follow-up (range, 3–60 months). However, given the small number of patients in this group, no conclusions on the effect of rituximab could be drawn. In a multicenter observational study of 4765 solid organ transplant recipients, antiviral prophylaxis was found to have no beneficial effect on early or late EBV+ PTLT occurrence. However, of the 191 patients who received rituximab-containing induction regimen, none developed PTLT. In contrast, 1.2% of patients (57/4574) who did not receive rituximab induction developed PTLT.⁵⁸ The 2019 AST-IDCOP guidelines recommend monitoring EBV viral load and preemptive interventions in patients who are EBV-seronegative pretransplant (weak/low).⁵⁵ Currently, EBV surveillance and preemptive therapy cannot be routinely recommended because of the lack of standardized time points for monitoring, cutoff values, and sources of samples (e.g., whole blood, plasma, serum, peripheral blood mononuclear cells, or isolated B cells) and the unclear risks and benefits of immunosuppression reduction. Nonetheless, this approach warrants further exploration.

Skin Cancer

Skin cancers are the most common *de novo* posttransplant tumors in the adult transplant population. Risk factors include light skin color, total sun burden and recreational sun exposure, genetic factors, history or present use of AZA, increasing number of immunosuppressants used, pretransplant history of squamous cell carcinoma (SCC), older age at transplant (>50 years), male sex, and duration of follow-up after transplantation. Cumulative UV radiation exposure has been suggested to be the primary carcinogen. Daily high-SPF sunscreen use is associated with a decrease in the incidence of actinic keratosis, SCC, and basal cell carcinoma (BCC) in solid organ transplant recipients. Potential pathogenic mechanisms of photocarcinogenesis include direct DNA damage, UV effects on host immunity, and synergistic effect with immunosuppressive drugs (particularly CNIs and AZA).⁵⁹ Limited studies suggest that AZA can also directly cause UVA-mediated DNA mutagenesis. Nonmelanoma skin cancer risk appears to be lower with MMF than with AZA use. In contrast, sirolimus or mTOR inhibitors may delay the onset or reduce the incidence of posttransplant skin and nonskin malignant neoplasms (discussed further later). Voriconazole (a previously commonly used antifungal agent to treat invasive fungal infection in organ transplantation) has been reported to be associated with a 73% increased risk of developing SCC. The newer antifungal agents posaconazole and isavuconazole have not been shown to increase skin cancer risk to date. Whereas BCC is the most common type of skin cancer in the general population, SCC has been reported to be 2 to 5 times more common than BCC in recipients of solid organ transplantation. In addition, immunosuppression in combination with enhanced sunlight exposure may induce malignant changes in papilloma-induced warts.

Immunosuppressive Therapy in Posttransplantation Malignancy

In principle, immunosuppression dose reduction improves immune surveillance against malignant cells. However, no systematic

studies have been performed to determine whether immunosuppression reduction or withdrawal benefits the natural history of established posttransplantation malignancy. In our opinion, a switch from CNI to sirolimus or CNI minimization in conjunction with sirolimus may be a viable therapeutic option in selected posttransplant malignancies (the antitumoral effect of sirolimus is discussed later). In patients with metastatic cancer, manipulation of immunosuppression is probably futile, and the risk for rejection and graft loss likely outweighs the benefit.

Immunosuppressive agents appear to have different effects on cancer risk after transplantation. The carcinogenic effects of antithymocyte globulin, cyclosporine, tacrolimus, and AZA have been well documented. In contrast to AZA, MFA derivatives have been shown to have an antiproliferative effect and have been suggested to be protective of posttransplantation malignancy. Analysis of more than 17,000 adult patients with preexisting diabetes indicated a significantly higher incidence of malignancy in AZA-treated than in MMF-treated patients (3.7% vs. 2.2%). In a nested case-control study to evaluate the association between immunosuppressive medications and squamous cell skin cancer risk among cardiac ($n = 35$) and kidney transplant recipients (kidney alone, $n = 124$; simultaneous kidney-pancreas, $n = 11$), an inverse association between MFA derivatives and squamous cell skin cancer was observed independent of tacrolimus use. MFA use was associated with a 57% squamous cell skin cancer risk reduction compared with no MFA derivative use. In contrast, AZA use was associated with a greater than twofold increase in squamous cell skin cancer risk compared with no AZA use.⁶⁰ In a study of 544 lung transplant recipients with a median survival of 11 years, the sequential use of AZA and MMF, each for at least 1 year, was similarly found to be associated with a lower risk of developing a cutaneous SCC compared with AZA use only (hazard ratio [HR], 0.24; 95% CI, 0.10–0.56).⁶¹ Limited studies suggest that AZA to MMF conversion therapy reduces skin photosensitivity to UVA and attenuates UVA-induced skin DNA damage. Nonetheless, one single-center study consisting of 930 heart transplant recipients demonstrated a lower incidence of skin cancer among patients receiving CsA/AZA combination therapy compared with those receiving CsA/MMF or Tac/MMF combination therapy.⁶² Whether MMF is protective of posttransplant malignancies remains unclear. Nonetheless, it is common practice to reduce or discontinue MMF in the presence of malignancies because of the theoretical risk for net protumor effect of overimmunosuppression.

Both sirolimus and everolimus have antiproliferative and antitumoral effects. mTOR inhibitor-containing regimens are associated with decreased nonmelanoma skin cancer risks, and their use is effective in both primary and secondary skin cancer prevention. The earlier the conversion after an initial diagnosis of cutaneous SCC, the greater the efficacy. The protective effect of mTOR inhibitors against skin cancer may be a result of its inhibition of several UV-induced mechanisms involved in skin carcinogenesis. The mTOR inhibitors are also effective in inducing remission of Kaposi sarcoma in organ transplant recipients.⁴⁴

Although sirolimus appears to have a beneficial effect for preventing primary and secondary skin cancer, its use in the management of other malignancies after solid organ transplantation remains to be defined. A systematic review of 20 RCTs and two observational studies demonstrated that sirolimus use was associated with lower overall cancer incidence, driven by a reduction in nonmelanoma skin cancer incidence. The protective effect of sirolimus on nonmelanoma skin cancer (NMSC) risk was most notable in studies comparing sirolimus against cyclosporine. After excluding NMSCs, there was no overall association between sirolimus and incidence of other cancers. Further analysis demonstrated that sirolimus use was associated with lower

kidney cancer incidence (IRR = 0.40; 95% CI, 0.20–0.81), and higher prostate incidence (IRR = 1.85; 95% CI, 1.17–2.91). The lower incidence of NMSC among sirolimus users was thought to be due in part to cyclosporine withdrawal. Overall, these findings suggested that sirolimus may reduce kidney cancer risk but is not protective for other cancers (including non-Hodgkin lymphoma, Kaposi sarcoma, and lung cancers) and may increase prostate cancer risk.⁶³ Notably, controlled trials demonstrated that although sirolimus was associated with a reduction in the risk of malignancy and nonmelanoma skin cancer in kidney transplant recipients, its use was associated with increased mortality risk compared with controls. This led some investigators to argue against mTOR inhibitor use in all patients with posttransplant malignancies.⁴⁸

We start patients with newly diagnosed skin cancer, renal cell carcinoma, or Kaposi sarcoma on an mTOR inhibitor in conjunction with discontinuation of antimetabolites and reduction of CNI therapy tailored to each individual patient. In those with a history of these malignancies prior to transplant, mTOR inhibitor therapy is generally not introduced until after the first 3 posttransplant months. There are insufficient data to recommend or refute its use in PTLD. Available data do not support the routine use of mTOR inhibitors in the management of other cancer types. Of interest, however, experimental mouse models demonstrated that everolimus significantly inhibits breast cancer cell growth, migration, and invasion.⁶⁴ The use of mTOR inhibitor in human breast cancer in the settings of transplantation warrants further studies.

SELF-ASSESSMENT QUESTIONS

- Which one of the following statements regarding immunization before and after transplantation is *correct*?
 - Seasonal influenza vaccine is contraindicated in the early post-transplant period because it is associated with allograft dysfunction and acute rejection.
 - COVID-19 vaccine is contraindicated in kidney transplant recipients with prior COVID-19 infection.
 - Live virus vaccination should be administered at least 2 weeks before transplantation to minimize the possibility of live vaccine-derived infection in the posttransplantation period.
 - Immunization of pets with live vaccines (such as canine *Bordetella bronchiseptica* intranasal vaccine) poses little to no risk of transmission.
- Clinical manifestations and management of CMV infections include all of the following *except*:
 - The absence of CMV DNAemia rules out invasive CMV disease.
 - Primary CMV infection is usually more severe than reactivated infection or superinfection.
 - Plasma CMV DNA should be monitored weekly during treatment.
 - The Third International Consensus guidelines on the management of CMV advocate foscarnet as the second-line agent for the treatment of ganciclovir-resistant CMV infection.
- Which of the following statements regarding posttransplantation lymphoproliferative disorder (PTLD) is *correct*?
 - Antiviral therapy with acyclovir or ganciclovir should be considered in Epstein-Barr virus-positive PTLD refractory to immunosuppression reduction alone.
 - Substitution of calcineurin inhibitor (cyclosporine or tacrolimus) for an mTOR inhibitor (sirolimus or everolimus) is considered standard of care.
 - Rituximab is considered as first-line treatment for CD20-positive PTLD that fails to respond to manipulation of immunosuppression alone.
 - PTLD is the most common type of posttransplantation malignancy in adult kidney transplant recipients.
 - A and C.

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